

Exact tangent visibility of finite causal-order laws and Fisher resolution in the S1 model

Abstract

Bombelli (2000)’s statistical geometry of finite causal-set laws gives us the object we study here: the unlabeled causal-order channel of an explicit $1 + 1$ -dimensional S1 interaction model, expanded at an independent reference geometry. For each fixed cardinality N , the score derivative factors through the orthogonal projection onto $V_N = \text{Sym}^2 P_{N-1}$, where P_{N-1} is generated by the first $N - 1$ centered shifted-Legendre modes, and this restriction to V_N is injective, so its kernel is the orthogonal symmetric complement of V_N together with the full antisymmetric sector. The spaces V_N have dimension $\binom{N}{2}$, are strictly nested, and their union is dense in the symmetric interaction Hilbert space. Visibility and statistical resolution turn out to be different questions: a positive-definite Fisher operator on V_N governs the latter, is generally anisotropic, and need not be diagonal in the natural modal basis, so visible modes can mix before they are resolved. We also exhibit an antisymmetric exponential orbit whose finite-poset law is even in the perturbation parameter, yet whose first nonzero jet is exactly of order two for every $N \geq 2$. This is a local differential resolution of Bombelli (2000)’s finite-law construction in S1, not a claim of nonlinear reconstruction or universal causal determination, and it does not extend beyond this model.

1 Introduction

Bombelli (2000) introduced the object this paper studies: the full law of an unlabeled causal poset at fixed cardinality N , sampled from a Lorentzian geometry, together with a statistical notion of closeness between two such laws. Janson (2011) supplies the general limiting framework of poset kernels and consistent finite laws in which this construction sits. Surya (2026) gives, via expected interval abundances, a closely related account of how increasing N can lift degeneracies in this kind of causal compression. None of the three computes the differential of the finite- N law at a reference geometry, its kernel, or its rank.

We work in an explicit $1 + 1$ -dimensional causal-diamond model (S1) with an independent-reference null geometry. The question we ask is narrow: to first order in a perturbation of the underlying Lorentzian geometry, what survives the map from a continuous perturbation to the finite unlabeled-poset law?

Two pieces of an answer already exist, in adjacent literatures, without being combined. At the level of *labeled* permutations rather than *unlabeled two-dimensional posets*, the differential of a permutation-pattern statistic around the uniform reference is understood in some detail. Even-Zohar (2020) decomposes the full space of pattern densities via the representation theory of S_N , isolating the standard-representation block V_1^{EZ} of dimension $(N - 1)^2$, realized explicitly via $U^T A(\sigma) U$. Kurečka (2022) differentiates the pattern density directly, expressing the gradient in a Bernstein-type basis on $E_N = \mathbf{1}^\perp$ via compressed permutation matrices $A_\pi|_{E_N}$ and covering-matrix sums $\sum_\pi t_\pi A_\pi$.

In our matrix convention, the distinction can be displayed as

$$\mathbb{R}^{\mathcal{C}_N} \xrightarrow{J_N} \mathbb{R}^{S_N} \xrightarrow{T_N} \text{End}(E_N), \quad J_N e_C = \mathbf{1}_{\Gamma_C}, \quad T_N(t) = \sum_{\pi \in S_N} t_\pi P_\pi \Big|_{E_N}. \quad (1.1)$$

Kurečka identifies the permutation-level differential through the second arrow: his gradient-polynomial map has kernel $\ker T_N$. The additional causal-order quotient restricts the domain to the fiber-constant subspace $\text{im } J_N$. The result proved here determines the image of that restriction, not merely the kernel of the ambient map.

The abstract target module is not new either. Diaconis (1989) and the monograph (Diaconis, 1988) give, for unordered-pair effects on rankings, the decomposition $M^{(N-2,2)} \simeq S^{(N)} \oplus S^{(N-1,1)} \oplus S^{(N-2,2)}$ — a module of dimension $\binom{N}{2}$ that is, up to naming, our target space $\text{Sym}^2 P_{N-1}$, illustrated already in 1988 with the Diallel Cross Design. And the many-to-one map from labeled permutations to unlabeled two-dimensional posets that we quotient by is itself classical combinatorics: Bayoumi et al. (1994) describe the fibers of this map explicitly, including their closure under $\sigma \mapsto \sigma^{-1}$ and the near-uniqueness of realizers for prime posets.

None of these five results takes the one further step that connects them: sum the permutation-level differential over the fibers Γ_C of the permutation-to-unlabeled-poset map, and show that the resulting class-sum score representatives *span* the full symmetric target module,

$$\text{span}\{R_C^{(N)} : C \in \mathcal{C}_N\} = \text{Sym}^2 P_{N-1}.$$

Equivalently, Theorem 1 (§4) proves the causal-order-specific identity

$$\boxed{T_N(\text{im } J_N) = \text{Sym}(E_N)}. \quad (1.2)$$

The exact kernel, the identifiable quotient, and the Fisher-resolution statement all turn on this span theorem. Corollaries 2 and 3 and Theorem 4 follow from it; Theorem 5 and Corollary 6 are a separate, self-contained second-order statement about one explicit antisymmetric orbit. Section 8 compares (1.1)–(1.2) directly with Kurečka’s covering-matrix technique.

Put narrowly, this is the contribution:

previous work identifies the ambient permutation-level structure;
we identify exactly what survives the quotient to unlabeled
finite causal-order laws.

We write "we identify" and "we prove," not "for the first time" or "novel." The individual ingredients above have substantial precedents, discussed in §8. We are not aware of this exact class-sum span theorem already stated in the literature, but that is not the same thing as a claim that none exists (§10).

We make no claim of nonlinear reconstruction, geometric identifiability at finite distance, or any result about Schwarzschild, horizons, or higher dimensions.

Our contributions are:

- C1** The exact identification of the all- N visible subspace $V_N = \text{Sym}^2 P_{N-1}$, together with its rank $\binom{N}{2}$, strict nesting, and symmetric density in the interaction Hilbert space. This is the paper’s principal mathematical contribution, established at first order in the S1 model at the independent reference geometry.

- C2** The exact operator factorization $D\mathcal{S}_N = B_N P_N^{\text{vis}}$, its kernel, and the identifiable quotient it induces, together with the resulting separation between which tangent directions are visible and how strongly the finite law resolves them through F_N . This is the paper’s principal operator-theoretic formulation; it is a functional corollary of the exact span in C1, not an independent new theory.
- C3** The asymptotic retention of Fisher information in the symmetric Hilbert–Schmidt sector, together with illustrative exact finite spectra. This is a quantitative resolution result, stated for an explicit channel and reference normalization.
- C4** The exact second-order detectability $r_N(\gamma_\psi) = 2$ for every $N \geq 2$ of one explicit antisymmetric orbit. This is a short, self-contained extension establishing structural existence, not a classification of the full antisymmetric sector $\bigwedge^2 H$.

2 The S1 model and finite causal-order experiments

We work in the flat $1 + 1$ -dimensional causal diamond in null coordinates, reparametrized to $D = [0, 1]^2$ with the product order $(u, v) \preceq (u', v') \iff u \leq u', v \leq v'$, and the uniform reference measure $\mu_0(du dv) = du dv$. A conformal generator $\psi \in C(D; \mathbb{R})$ defines the normalized exponential family

$$g_\varepsilon = \frac{e^{2\varepsilon\psi}}{Z(\varepsilon)} g_0, \quad Z(\varepsilon) = \int_D e^{2\varepsilon\psi} d\mu_0,$$

which preserves total volume at every ε — not only to first order — and induces the sampling density $q_\varepsilon = e^{2\varepsilon\psi}/Z(\varepsilon)$ for a Poisson sprinkling conditioned on N points. Differentiating at $\varepsilon = 0$ gives the metric tangent $\dot{g}_0 = 2(\psi - \bar{\psi})g_0$, with $\bar{\psi}$ the mean of ψ under μ_0 .

Four distinct objects appear along the chain from geometry to statistic, and we keep them notationally separate throughout: the geometric generator ψ ; the log-tangent of the normalized joint density $t_\psi = 2(\psi - \bar{\psi})$; the copula-density tangent h_ψ , obtained after uniformizing both marginals,

$$h_\psi(u, v) = 2[\psi(u, v) - \psi_U(u) - \psi_V(v) + \bar{\psi}] = 2\mathcal{P}\psi, \quad \mathcal{P} = (I - M_u)(I - M_v),$$

with ψ_U, ψ_V the two marginal means of ψ and $\mathcal{P}$ the double-centering projection used throughout §§4–6; and the score $S_{N,\psi}$ of the finite discrete experiment, defined below. The joint and copula densities both equal 1 at $\varepsilon = 0$, so t_ψ and h_ψ agree there in value but are not the same object — they differ by the two marginal terms, and only h_ψ is what the finite experiment actually sees, since passing to ranks applies the marginal probability integral transform and removes marginal information.

A sample of N points generates three progressively coarser observations. At the finest level is the continuous sample $(U_k, V_k)_{k=1}^N$ itself, with score $T_{N,\psi} = \sum_k h_\psi(U_k, V_k)$. Ordering by U and recording the induced rank in V produces a *labeled* rank permutation $\Pi_N \in S_N$; its score is the conditional expectation

$$S_{N,\psi}(\pi) = \mathbb{E}_0[T_{N,\psi} \mid \Pi_N = \pi],$$

an identity that needs no independence assumption after conditioning and follows directly from the likelihood. Finally — and this is the level the finite causal-order law actually lives on — the *unlabeled* two-dimensional poset is the isomorphism class $[P_{\Pi_N}]$ of the permutation matrix: Π_N depends on which linear realizer of U and of V is used, but the order-only datum a causal set exposes is exactly this isomorphism class, invariant under the realizer. The quotient from the labeled

permutation Π_N to the unlabeled poset $[P_{\Pi_N}]$ — and exactly what survives it — is the subject of §§4–5.

Write $\mathcal{C}_N$ for the set of isomorphism classes of two-dimensional posets realized by some $\sigma \in S_N$, and for $C \in \mathcal{C}_N$ let $\Gamma_C := \{\sigma \in S_N : [P_\sigma] = C\}$ be its fiber under $\sigma \mapsto [P_\sigma]$. The object this paper’s tangent statement is about is the law of the unlabeled poset itself,

$$\mu_{N,\varepsilon}^{[P]}(C) := \mathbb{P}_\varepsilon([P_{\Pi_N}] = C) = \sum_{\sigma \in \Gamma_C} p_\varepsilon(\sigma), \quad C \in \mathcal{C}_N,$$

with reference value $\mu_{N,0}(C) = |\Gamma_C|/N!$. This closes the chain the rest of the paper works with:

$$\psi \longrightarrow \dot{g}_0 \longrightarrow t_\psi \longrightarrow h_\psi \longrightarrow S_{N,\psi} \longrightarrow \mu_{N,\varepsilon}^{[P]}.$$

Throughout, $H = L_0^2([0, 1])$ denotes the mean-zero L^2 space with the shifted-Legendre basis, $P_{N-1} = \text{span}\{p_1, \dots, p_{N-1}\} \subset H$ its first $N - 1$ modes, and derivatives of finite- N laws at $\varepsilon = 0$ are read as scores under the quadratic-mean differentiability (QMD) identification, specializing Bombelli (2000)’s finite-law construction to this S1 exponential family.

3 Score representatives of the finite law

For $i = 1, \dots, N$, let

$$d_i^{(N)}(t) := N \binom{N-1}{i-1} t^{i-1} (1-t)^{N-i} \quad (3.1)$$

be the density of the i -th order statistic in a sample of N independent uniform variables, and define its centered version

$$b_i^{(N)} := d_i^{(N)} - 1. \quad (3.2)$$

The normalized functions $d_i^{(N)}/N$ form the Bernstein basis of degree $N - 1$. Since

$$\sum_{i=1}^N d_i^{(N)} = N, \quad \sum_{i=1}^N b_i^{(N)} = 0,$$

centering leaves exactly one linear relation and therefore

$$\text{span}\{b_1^{(N)}, \dots, b_N^{(N)}\} = P_{N-1}. \quad (3.3)$$

Let $f = \mathcal{P}\psi \in H \widehat{\otimes} H$ be an admissible S1 interaction tangent, so the score of one observation of the copula at the reference model is $2f$. If $p_\varepsilon(\sigma)$ denotes the probability of the rank permutation $\sigma \in S_N$, differentiation of its finite likelihood gives

$$p'_\sigma(0; f) = \frac{2}{N!} \sum_{i=1}^N \left\langle f, d_i^{(N)} \otimes d_{\sigma(i)}^{(N)} \right\rangle. \quad (3.4)$$

Both marginals of f vanish. Hence each $d_i^{(N)}$ in (3.4) may be replaced by $b_i^{(N)}$, and the permutation-level representative is defined by

$$\boxed{R_\sigma^{(N)} := \frac{2}{N!} \sum_{i=1}^N b_i^{(N)} \otimes b_{\sigma(i)}^{(N)}, \quad p'_\sigma(0; f) = \langle f, R_\sigma^{(N)} \rangle.} \quad (3.5)$$

In particular, $R_\sigma^{(N)} \in P_{N-1} \otimes P_{N-1}$. Formula (3.5), initially obtained for admissible continuous tangents, also defines a continuous linear functional of every $f \in H \widehat{\otimes} H$; below we use this Hilbert-space extension without claiming that every such f is geometrically realizable.

Recall from §2 that $\mathcal{C}_N$ is the set of unlabeled two-dimensional poset classes generated at cardinality N , and $\Gamma_C = \{\sigma \in S_N : [P_\sigma] = C\}$ is the fiber of $C \in \mathcal{C}_N$. The class representative is

$$R_C^{(N)} := \sum_{\sigma \in \Gamma_C} R_\sigma^{(N)}. \quad (3.6)$$

Since $\mu_{N,0}(C) = |\Gamma_C|/N! > 0$, summing (3.5) over the fiber gives

$$\left. \frac{d}{d\varepsilon} \mu_{N,\varepsilon}^{[P]}(C) \right|_{\varepsilon=0} = \langle f, R_C^{(N)} \rangle, \quad (D\mathcal{S}_N f)(C) = \frac{\langle f, R_C^{(N)} \rangle}{\mu_{N,0}(C)}. \quad (3.7)$$

Thus $D\mathcal{S}_N f$ is exactly the score of the finite unlabeled-poset law. Its $\mu_{N,0}$ -mean is zero because the derivatives in (3.7) sum to zero over C .

Each fiber is closed under inversion: if $i < j$ then $\sigma(i) < \sigma(j)$, so setting $a = \sigma(i)$, $b = \sigma(j)$ gives $a < b$ and $\sigma^{-1}(a) = i < j = \sigma^{-1}(b)$, i.e. $i \mapsto \sigma(i)$ is a poset isomorphism $P_{\sigma^{-1}} \cong P_\sigma$, so interchanging the two rank coordinates sends σ to σ^{-1} without changing the abstract poset. Consequently,

$$\sigma \in \Gamma_C \iff \sigma^{-1} \in \Gamma_C, \quad \left(R_\sigma^{(N)}\right)^\top = R_{\sigma^{-1}}^{(N)},$$

and hence

$$\left(R_C^{(N)}\right)^\top = R_C^{(N)}, \quad R_C^{(N)} \in \text{Sym}^2 P_{N-1}. \quad (3.8)$$

The Fisher bilinear form of the unlabeled-poset experiment is therefore defined by

$$G_{[P]}^{(N)}(f, g) := \langle D\mathcal{S}_N f, D\mathcal{S}_N g \rangle_{L^2(\mu_{N,0})} = \sum_{C \in \mathcal{C}_N} \frac{\langle f, R_C^{(N)} \rangle \langle g, R_C^{(N)} \rangle}{\mu_{N,0}(C)}. \quad (3.9)$$

Because every reference mass is strictly positive,

$$\ker D\mathcal{S}_N = \ker G_{[P]}^{(N)} = \text{span}\{R_C^{(N)} : C \in \mathcal{C}_N\}^\perp,$$

and the visible space is equivalently

$$V_N := (\ker D\mathcal{S}_N)^\perp = \text{span}\{R_C^{(N)} : C \in \mathcal{C}_N\} \subseteq \text{Sym}^2 P_{N-1}. \quad (3.10)$$

This is the upper inclusion used in §4.

To state the remaining finite reduction, put

$$E_N := \mathbf{1}^\perp \subset \mathbb{R}^N, \quad \Lambda_N : E_N \longrightarrow P_{N-1}, \quad \Lambda_N z := \sum_{i=1}^N z_i b_i^{(N)}. \quad (3.11)$$

Equation (3.3) and the single relation $\sum_i b_i^{(N)} = 0$ show that Λ_N is an isomorphism. Fix the permutation-matrix convention

$$P_\sigma := \sum_{i=1}^N e_i e_{\sigma(i)}^\top, \quad A_C := \sum_{\sigma \in \Gamma_C} P_\sigma. \quad (3.12)$$

Closure of Γ_C under inversion makes A_C symmetric, and (3.5)–(3.6) give exactly

$$R_C^{(N)} = \frac{2}{N!} \sum_{i,j=1}^N (A_C)_{ij} b_i^{(N)} \otimes b_j^{(N)}. \quad (3.13)$$

Transporting by the isomorphism Λ_N therefore reduces the reverse inclusion in (3.10) to

$$\boxed{V_N = \text{Sym}^2 P_{N-1} \iff \text{span}\{A_C|_{E_N} : C \in \mathcal{C}_N\} = \text{Sym}(E_N).} \quad (3.14)$$

Section 4 proves the right-hand statement constructively for every $N \geq 2$. The Bernstein and permutation-level differential machinery in this section is not claimed as new; the all- N contribution isolated in §4 is the span that survives after summing over the unlabeled-poset fibers.

4 Exact all- N visible subspaces

By (3.10), the visible space $V_N = \text{span}\{R_C^{(N)} : C \in \mathcal{C}_N\}$ satisfies $V_N \subseteq \text{Sym}^2 P_{N-1}$. Under the isomorphism Λ_N of (3.11), equation (3.14) reduces the reverse inclusion to the following purely combinatorial statement about the class sums defined in (3.12):

$$\text{span}\{A_C|_{E_N} : C \in \mathcal{C}_N\} = \text{Sym}(E_N). \quad (4.1)$$

Theorem 1 (class-sum span). *For every $N \geq 2$,*

$$\boxed{V_N = \text{Sym}^2 P_{N-1}, \quad \dim V_N = \text{rank } G_{[P]}^{(N)} = \binom{N}{2}.$$

Equivalently, $T_N(\text{im } J_N) = \text{Sym}(E_N)$. The proof is constructive at every N : it exhibits an explicit family of $\binom{N}{2}$ poset classes whose score representatives span the target space, without enumerating posets or extrapolating from small N .

Proof. We establish (4.1) in three steps.

A family of $\binom{N}{2}$ near-chain posets. For integers $0 \leq a < b \leq N-1$, define $C_{a,b}$ on a chain $c_1 < \dots < c_{N-1}$ together with one extra element z satisfying $c_i < z$ for $i \leq a$, $z < c_i$ for $i > b$, and z incomparable with $c_{a+1}, \dots, c_b$. Every linear extension inserts z after exactly $k \in \{a, \dots, b\}$ elements of the chain, and two extensions realize the same order relation exactly when $\{s, t\} = \{a, b\}$: the poset has exactly two linear-extension pairs, giving, once the chain is normalized to $1 < \dots < N$, a single cycle $\tau_{a,b}$ on the consecutive interval $I_{a,b} = \{a+1, \dots, b+1\}$ and its inverse. Hence

$$\Gamma_{C_{a,b}} = \{\tau_{a,b}, \tau_{a,b}^{-1}\}$$

(a single transposition, without multiplicity, when $b = a+1$). The multiset of strict-past cardinalities $\{|\text{Past}(y)| : y \in C_{a,b}\} = \{0, \dots, b-1, b+1, \dots, N-1\} \uplus \{a\}$ is missing b and repeats a , so it determines the pair (a, b) : the $\binom{N}{2}$ classes $C_{a,b}$ are pairwise distinct.

From interval cycles to edge Laplacians. For $1 \leq i < j \leq N$ let $L_{ij} := (e_i - e_j)(e_i - e_j)^\top$ be the Laplacian of edge $\{i, j\}$ in the complete graph on N vertices. Restricted to E_N , the $\binom{N}{2}$ matrices $\{L_{ij}\}$ are linearly independent — a vanishing combination forces every off-diagonal coefficient, hence every w_{ij} , to vanish — so they form a basis of $\text{Sym}(E_N)$; they also satisfy $\sum_{i < j} L_{ij} = NI_{E_N}$. Set $S_{a,b} := P_{\tau_{a,b}} + P_{\tau_{a,b}}^\top$ — equal to $2A_{C_{a,b}}$ when $b = a+1$ and to $A_{C_{a,b}}$ otherwise, a nonzero scalar either way — and $Q_{a,b} := 2I_{E_N} - S_{a,b}|_{E_N}$. Because $\tau_{a,b}$ is the consecutive cycle on $I_{a,b}$, $Q_{a,b}$ is exactly that

cycle's graph Laplacian: $Q_{a,a+1} = 2L_{a+1,a+2}$, and for $b > a + 1$, $Q_{a,b} = L_{a+1,b+1} + \sum_{k=a+1}^b L_{k,k+1}$. These two identities triangularize by interval length and invert cleanly — $L_{i,i+1} = \frac{1}{2}Q_{i-1,i}$, and for $j > i + 1$, $L_{ij} = Q_{i-1,j-1} - \frac{1}{2} \sum_{k=i}^{j-1} Q_{k-1,k}$ — so

$$\text{span}\{Q_{a,b}\} = \text{span}\{L_{ij} : i < j\} = \text{Sym}(E_N). \quad (4.2)$$

This leaves exactly one gap: (4.2) uses the shared term $2I_{E_N}$ subtracted out of every $Q_{a,b}$, so it does not by itself show that the class sums $S_{a,b}$ span the same space.

Closing the gap: the identity is itself a class-sum combination. From $\sum_{i < j} L_{ij} = NI_{E_N}$ and the triangularized identities above, $I_{E_N} = \sum_{a < b} c_{a,b} Q_{a,b}$ for explicit coefficients $c_{a,b}$ whose sum depends only on how many edges sit at each distance $d = j - i$:

$$s_N := \sum_{a < b} c_{a,b} = \frac{1}{N} \sum_{d=1}^{N-1} (N-d) \left(1 - \frac{d}{2}\right) = \frac{(N-1)(5-N)}{12}.$$

Substituting $Q_{a,b} = 2I_{E_N} - S_{a,b}|_{E_N}$ gives $(1 - 2s_N)I_{E_N} = -\sum_{a < b} c_{a,b} S_{a,b}|_{E_N}$, and the coefficient

$$1 - 2s_N = \frac{N^2 - 6N + 11}{6} = \frac{(N-3)^2 + 2}{6}$$

is strictly positive for every integer N — it never vanishes, at $N = 3$ or anywhere else — so $I_{E_N} \in \text{span}\{S_{a,b}|_{E_N}\}$. Feeding this back into $Q_{a,b} = 2I_{E_N} - S_{a,b}|_{E_N}$ shows every $Q_{a,b}$ is itself in that span, and with (4.2),

$$\text{span}\{S_{a,b}|_{E_N} : a < b\} = \text{Sym}(E_N).$$

Since each $S_{a,b}$ is $A_{C_{a,b}}$ up to a nonzero scalar, this is exactly (4.1). Combined with §3's reduction, this proves Theorem 1. $\square$

Corollary 2 (filtration). Since $P_{N-1} \subsetneq P_N$,

$$V_N = \text{Sym}^2 P_{N-1} \subsetneq \text{Sym}^2 P_N = V_{N+1} \quad (N \geq 2),$$

with rank sequence $1, 3, 6, 10, 15, \dots, \binom{N}{2}, \dots$ — a strict inclusion, not merely non-decreasing, at every step. A universal witness of each new degree is $p_1 \otimes p_N + p_N \otimes p_1 \in V_{N+1} \setminus V_N$: since the Fisher form is positive definite on V_{N+1} and vanishes on $V_N^\perp$, $I_N^{[P]}(p_1 \otimes p_N) = 0$ while $I_{N+1}^{[P]}(p_1 \otimes p_N) > 0$ — the same witness recurs in the Fisher computations of §6. Because centered polynomials are dense in $H = L_0^2([0, 1])$, Theorem 1 also gives, unconditionally,

$$\overline{\bigcup_{N \geq 2} V_N} = H \widehat{\otimes}_{\text{sym}} H.$$

The exact kernel of $D\mathcal{S}_N$ — the orthogonal complement of V_N together with the permanently invisible antisymmetric sector $\bigwedge^2 H$ — is not derived here; it is Corollary 3, §5, the direct functional consequence of Theorem 1 once the span is known.

The finite reduction used in the argument is recorded in Appendix B, while Appendix C collects the almost-chain and Laplacian construction in full detail.

5 Operator factorization and the identifiable quotient

Let

$$\mathcal{X} := H \widehat{\otimes} H, \quad \mathcal{K}_N := L_0^2(\mathcal{C}_N, \mu_{N,0}),$$

where $\mathcal{K}_N$ is the Hilbert space of mean-zero scores of the finite unlabeled-poset law at the reference model. Under the QMD identification of §2, write

$$D\mathcal{S}_N : \mathcal{X} \longrightarrow \mathcal{K}_N$$

for its bounded score differential. The visible subspace is, by definition,

$$V_N = (\ker D\mathcal{S}_N)^\perp. \quad (5.1)$$

Theorem 1 identifies this abstract support exactly:

$$V_N = \text{Sym}^2 P_{N-1}. \quad (5.2)$$

Denote by Π_{sym} the orthogonal projection of $\mathcal{X}$ onto $H^{\widehat{\otimes}_{\text{sym}}} H$, by Π_{V_N} the projection of the latter space onto V_N , and set

$$P_N^{\text{vis}} := \Pi_{V_N} \Pi_{\text{sym}}. \quad (5.3)$$

Thus

$$\mathcal{X} = V_N \oplus V_N^{\perp_{\text{sym}}} \oplus \bigwedge^2 H, \quad \ker P_N^{\text{vis}} = V_N^{\perp_{\text{sym}}} \oplus \bigwedge^2 H. \quad (5.4)$$

Corollary 3 (exact factorization, kernel, and identifiable quotient). *For every $N \geq 2$, let*

$$B_N := D\mathcal{S}_N|_{V_N} : V_N \longrightarrow \mathcal{K}_N.$$

Then B_N is injective and

$$\boxed{D\mathcal{S}_N = B_N P_N^{\text{vis}}}, \quad (5.5)$$

$$\boxed{\ker D\mathcal{S}_N = V_N^{\perp_{\text{sym}}} \oplus \bigwedge^2 H = (\text{Sym}^2 P_{N-1})^{\perp_{\text{sym}}} \oplus \bigwedge^2 H}. \quad (5.6)$$

If $q_N : \mathcal{X} \rightarrow \mathcal{X}/\ker D\mathcal{S}_N$ is the quotient map, then

$$U_N : \mathcal{X}/\ker D\mathcal{S}_N \longrightarrow V_N, \quad U_N([f]) = P_N^{\text{vis}} f, \quad (5.7)$$

is a canonical isometric isomorphism for the Hilbert quotient norm, and the differential has the unique induced factorization

$$\mathcal{X} \xrightarrow{q_N} \mathcal{X}/\ker D\mathcal{S}_N \xrightarrow{U_N} V_N \xrightarrow{B_N} \mathcal{K}_N, \quad D\mathcal{S}_N = B_N U_N q_N. \quad (5.8)$$

In particular,

$$\boxed{\mathcal{X}/\ker D\mathcal{S}_N \simeq V_N = \text{Sym}^2 P_{N-1}, \quad \dim(\mathcal{X}/\ker D\mathcal{S}_N) = \binom{N}{2}}. \quad (5.9)$$

Proof. The orthogonal decomposition $\mathcal{X} = (\ker D\mathcal{S}_N)^\perp \oplus \ker D\mathcal{S}_N$ gives $D\mathcal{S}_N f = D\mathcal{S}_N P_N^{\text{vis}} f$, which is (5.5). The restriction B_N is injective because its domain is $(\ker D\mathcal{S}_N)^\perp$. Equations (5.2) and (5.4) then give (5.6). Finally, (5.6) makes (5.7) well defined and bijective, while

$$\|[f]\|_{\mathcal{X}/\ker D\mathcal{S}_N} = \inf_{k \in \ker D\mathcal{S}_N} \|f + k\|_{\mathcal{X}} = \|P_N^{\text{vis}} f\|_{\mathcal{X}};$$

hence U_N is isometric and (5.8) follows. $\square$

In particular, for two interaction tangents $f, g \in \mathcal{X}$,

$$D\mathcal{S}_N f = D\mathcal{S}_N g \iff [f] = [g] \text{ in } \mathcal{X} / \ker D\mathcal{S}_N \iff P_N^{\text{vis}} f = P_N^{\text{vis}} g. \quad (5.10)$$

This is the precise sense in which the finite law identifies the quotient: it is a statement about its first differential at the reference model, not nonlinear identifiability of geometries at finite distance.

Appendix D collects the kernel, quotient, strict-nesting and density arguments used here.

The projection P_N^{vis} specifies **which** tangent directions survive; it does not specify how strongly the finite law encodes different surviving directions. That second datum is

$$F_N := B_N^* B_N : V_N \longrightarrow V_N, \quad D\mathcal{S}_N^* D\mathcal{S}_N = P_N^{\text{vis}} F_N P_N^{\text{vis}}. \quad (5.11)$$

Since B_N is injective and V_N is finite dimensional, F_N is positive definite on V_N ; it need not be the identity and must not be conflated with the support projection. Its spectrum and anisotropy belong to §6.

6 Fisher resolution inside the visible sector

For $f \in \mathcal{X} = H \hat{\otimes} H$, the Fisher information in the finite unlabeled-poset law is

$$I_N^{[P]}(f) := G_{[P]}^{(N)}(f, f) = \|D\mathcal{S}_N f\|_{L^2(\mu_{N,0})}^2 = \langle P_N^{\text{vis}} f, F_N P_N^{\text{vis}} f \rangle. \quad (6.1)$$

The continuous reference experiment consists of N independent copula observations, whose score is $2 \sum_{k=1}^N f(U_k, V_k)$. Its Fisher form is therefore $4N \langle f, g \rangle$. We use it only as an explicit normalization and set

$$\hat{F}_N := \frac{1}{4N} P_N^{\text{vis}} F_N P_N^{\text{vis}} = \frac{1}{4N} D\mathcal{S}_N^* D\mathcal{S}_N \quad \text{on } \mathcal{X}. \quad (6.2)$$

By data processing and Corollary 3,

$$0 \leq \hat{F}_N \leq I_{\mathcal{X}}, \quad \text{supp } \hat{F}_N = V_N, \quad \ker \hat{F}_N = \ker D\mathcal{S}_N. \quad (6.3)$$

Thus P_N^{vis} fixes the support, whereas the nonzero eigenvalues of $\hat{F}_N$ quantify the resolution inside that support. In general

$$\hat{F}_N \neq P_N^{\text{vis}} \quad \text{and, equivalently,} \quad F_N \neq 4N I_{V_N}. \quad (6.4)$$

Exact low-cardinality spectra. For compact rational formulas put

$$x(t) := t - \frac{1}{2}, \quad q(t) := \left(t - \frac{1}{2}\right)^2 - \frac{1}{12}, \quad r(t) := \left(t - \frac{1}{2}\right)^3 - \frac{3}{20} \left(t - \frac{1}{2}\right). \quad (6.5)$$

These are mutually orthogonal and span P_1, P_2, P_3 successively. For $N = 2$, with $e_{11} = x \otimes x$,

$$G_{[P]}^{(2)}(f, g) = 256 \langle f, e_{11} \rangle \langle g, e_{11} \rangle, \quad \text{spec}_+(\hat{F}_2) = \left\{ \frac{2}{9} \right\}. \quad (6.6)$$

For $N = 3$, in the orthogonal basis

$$e_{11} = x \otimes x, \quad e_{12} = x \otimes q + q \otimes x, \quad e_{22} = q \otimes q,$$

the poset and continuous-reference forms are respectively

$$[G_{[P]}^{(3)}] = \text{diag} \left(\frac{1}{32}, \frac{1}{1200}, \frac{1}{180000} \right), \quad [G_{\text{full}}^{(3)}] = \text{diag} \left(\frac{1}{12}, \frac{1}{90}, \frac{1}{2700} \right). \quad (6.7)$$

Hence

$$\boxed{\text{spec}_+(\hat{F}_3) = \left\{ \frac{3}{8}, \frac{3}{40}, \frac{3}{200} \right\}}, \quad (6.8)$$

with eigenvectors e_{11}, e_{12}, e_{22} , in that order. The visible space is already anisotropic: support membership alone does not determine retained Fisher strength.

At $N = 4$, use

$$\begin{aligned} e_{11} &:= x \otimes x, & e_{12} &:= x \otimes q + q \otimes x, & e_{13} &:= x \otimes r + r \otimes x, \\ e_{22} &:= q \otimes q, & e_{23} &:= q \otimes r + r \otimes q, & e_{33} &:= r \otimes r. \end{aligned} \quad (6.9)$$

Three generalized eigenvectors remain pure:

$$\hat{F}_4 e_{11} = \frac{12}{25} e_{11}, \quad \hat{F}_4 e_{12} = \frac{4}{25} e_{12}, \quad \hat{F}_4 e_{23} = \frac{4}{525} e_{23}. \quad (6.10)$$

The remaining invariant block is $\text{span}\{e_{13}, e_{22}, e_{33}\}$. In that ordered basis, the restrictions of the two bilinear forms are

$$\begin{aligned} [G_{[P]}^{(4)}]_{\text{mix}} &= \begin{pmatrix} 1/55125 & 1/354375 & 1/38587500 \\ 1/354375 & 11/455625 & -1/49612500 \\ 1/38587500 & -1/49612500 & 11/5402250000 \end{pmatrix}, \\ [G_{\text{full}}^{(4)}]_{\text{mix}} &= \text{diag} \left(\frac{1}{1050}, \frac{1}{2025}, \frac{1}{490000} \right). \end{aligned} \quad (6.11)$$

Its three generalized eigenvalues are the real positive roots of

$$144703125\lambda^3 - 9975000\lambda^2 + 142000\lambda - 128 = 0, \quad (6.12)$$

namely, for orientation only,

$$0.0494521212879\dots, \quad 0.0185160720400\dots, \quad 0.000966047034941\dots$$

Therefore the exact spectrum is specified by (6.10) and (6.12), and its decreasing numerical order is

$$\frac{12}{25} > \frac{4}{25} > 0.0494521212879\dots > 0.0185160720400\dots > \frac{4}{525} > 0.000966047034941\dots > 0. \quad (6.13)$$

The off-diagonal entries in (6.11) are the first explicit modal mixing: e_{13}, e_{22}, e_{33} are visible but are not individually Fisher eigenvectors.

Theorem 4 (Fisher resolution and asymptotic retention). Let $S_N^\Pi(f)$ be the score before the quotient, when the full rank permutation Π_N is observed, and put

$$I_N^\Pi(f) := \mathbb{E}_0[S_N^\Pi(f)^2], \quad \Delta_N(f) := I_N^\Pi(f) - I_N^{[P]}(f). \quad (6.14)$$

Conditional expectation of scores along $\Pi_N \mapsto [P_{\Pi_N}]$ gives

$$\Delta_N(f) = \mathbb{E}_0[\text{Var}_0(S_N^\Pi(f) \mid [P_{\Pi_N}])] \geq 0. \quad (6.15)$$

For every $f \in \mathcal{X}$,

$$\frac{I_N^\Pi(f)}{N} \longrightarrow 4\|f\|_{\mathcal{X}}^2. \quad (6.16)$$

If in addition $f \in H \widehat{\otimes}_{\text{sym}} H$, then

$$\frac{\Delta_N(f)}{N} \longrightarrow 0, \quad \frac{I_N^{[P]}(f)}{N} \longrightarrow 4\|f\|_{\mathcal{X}}^2. \quad (6.17)$$

For every nonzero symmetric f , (6.16) gives a finite, generally nonuniform threshold $N_0(f)$ such that $I_N^\Pi(f) > 0$ for all $N \geq N_0(f)$. On that range define the two stages and their product by the following notation; κ_N is used for the second stage because q_N already denotes the quotient map in Corollary 3:

$$\rho_N(f) := \frac{I_N^\Pi(f)}{4N\|f\|^2}, \quad \kappa_N(f) := \frac{I_N^{[P]}(f)}{I_N^\Pi(f)}, \quad \eta_N^{\text{tot}}(f) := \rho_N(f)\kappa_N(f) = \frac{I_N^{[P]}(f)}{4N\|f\|^2}. \quad (6.18)$$

Then

$$\boxed{\rho_N(f) \longrightarrow 1, \quad \kappa_N(f) \longrightarrow 1, \quad \eta_N^{\text{tot}}(f) \longrightarrow 1.} \quad (6.19)$$

The first limit concerns continuous observations versus ranks; the second concerns ranks versus the unlabeled poset. They are distinct claims and neither is inferred from the strict inclusions $V_N \subsetneq V_{N+1}$.

Equivalently, on the full interaction Hilbert space,

$$\boxed{\widehat{F}_N \xrightarrow{\text{SOT}} \Pi_{\text{sym}}.} \quad (6.20)$$

More generally, if $0 \neq f = f_s + f_a$ is its orthogonal symmetric–antisymmetric decomposition, then

$$\frac{I_N^{[P]}(f)}{4N\|f\|^2} \longrightarrow \frac{\|f_s\|^2}{\|f_s\|^2 + \|f_a\|^2}. \quad (6.21)$$

In particular, the antisymmetric sector contributes zero to the numerator at every N , not merely asymptotically.

Proof. Define

$$H_{ij}^{(N)}(f) := \iint f(s, t) d_i^{(N)}(s) d_j^{(N)}(t) ds dt. \quad (6.22)$$

Its row and column sums vanish for $f \in \mathcal{X}$, and the permutation score is $S_N^\Pi(f) = 2 \sum_i H_{i, \Pi_N(i)}^{(N)}(f)$. Direct averaging over a uniform permutation gives the exact Gram identity

$$I_N^\Pi(f) = \frac{4}{N-1} \|H^{(N)}(f)\|_F^2. \quad (6.23)$$

The normalized order-statistic operator has the shifted-Legendre eigenvalues

$$\lambda_{N-1, m} = \prod_{r=1}^m \frac{N-r}{N+r} \in [0, 1], \quad \lambda_{N-1, m} \longrightarrow 1 \quad (m \text{ fixed}). \quad (6.24)$$

Expanding f in the tensor Legendre basis and applying dominated convergence to (6.23) proves (6.16). For symmetric finite-rank tensors, the fiber theorem gives $\Delta_N(f)/N \rightarrow 0$. The uniform bound

$$0 \leq \frac{\Delta_N(f)}{N} \leq 8\|f\|_{\mathcal{X}}^2 \quad (6.25)$$

extends this limit to the full symmetric Hilbert–Schmidt closure by a fixed-rank approximation followed by $N \rightarrow \infty$, proving (6.17). Equations (6.18)–(6.19) follow by division only after the

denominator is positive. Inversion symmetry annihilates the antisymmetric conditional score on every poset fiber, giving (6.21). Finally, (6.17), (6.21), polarization and $0 \leq \widehat{F}_N \leq I$ imply (6.20). $\square$

The convergence in (6.20) is not convergence in operator norm. Indeed, for the unit vector $h_N := p_N \otimes p_N / \|p_N\|_{L^2}^2$, Theorem 1 gives $h_N \perp V_N$, so

$$\widehat{F}_N h_N = 0, \quad \Pi_{\text{sym}} h_N = h_N, \quad \|\widehat{F}_N - \Pi_{\text{sym}}\| \geq 1 \quad \text{for every } N. \quad (6.26)$$

Nor does (6.20) make $\widehat{F}_N$ a finite- N projection: already (6.6) gives its nonzero eigenvalue $2/9$. The generic rate in (6.17)–(6.21) is only $o_f(1)$, with no rate or threshold uniform over the Hilbert–Schmidt unit sphere. The available $1 - \kappa_N(f) = O(N^{-1/2})$ rate applies only to the bounded continuous finite-rank subclass.

These statements concern the Hilbert completion of S1 interaction tangents. They transfer directly to the admissible geometric paths already constructed in that class, but they do not assert geometric realizability of an arbitrary Hilbert–Schmidt tensor. The finite spectra above compare the unlabeled-poset law with N continuous copula observations; they are not fractions of the full geometry, do not imply reconstruction, and do not establish monotonicity of individual Fisher eigenvalues with N .

Appendix E makes the retention step explicit: (E.15) identifies the high-probability fiber event, (E.16) proves the finite-rank loss bound, and (E.17)–(E.19) pass to the full symmetric Hilbert–Schmidt closure. Appendix F records the exact matrices and generalized spectra for $N = 2, 3, 4$.

7 An antisymmetric orbit visible at second order

Let $\ell_1, \ell_2 \in H$ be the first two orthonormal shifted-Legendre modes,

$$\ell_1(t) = \sqrt{3}(2t - 1), \quad \ell_2(t) = \sqrt{5}(6t^2 - 6t + 1),$$

and define the polynomial generator

$$\boxed{\psi(u, v) := \ell_1(u)\ell_2(v) - \ell_2(u)\ell_1(v).} \quad (7.1)$$

It is antisymmetric under coordinate exchange, has zero marginals, and satisfies

$$\mathcal{P}\psi = \psi \neq 0, \quad h_\psi = 2\psi \in \bigwedge^2 H, \quad \|\psi\|_{L^2(D)}^2 = 2, \quad \|h_\psi\|_{L^2(D)}^2 = 8. \quad (7.2)$$

In particular, $\psi \notin \ker \mathcal{P}$: its first-order invisibility is not the marginal gauge of §2. Since ψ is bounded, the normalized exponential family

$$\gamma_\psi : \varepsilon \mapsto g_\varepsilon = \frac{e^{2\varepsilon\psi}}{Z(\varepsilon)} g_0, \quad Z(\varepsilon) = \int_D e^{2\varepsilon\psi} d\mu_0, \quad (7.3)$$

is an admissible S1 path for every real ε .

Let $\iota : D \rightarrow D$ exchange the two null coordinates, $\iota(u, v) = (v, u)$. It preserves the product order, the reference measure and the flat metric, while $\psi \circ \iota = -\psi$. A change of variables also gives $Z(-\varepsilon) = Z(\varepsilon)$, and therefore

$$\boxed{\iota^* g_\varepsilon = g_{-\varepsilon}} \quad (\varepsilon \in \mathbb{R}). \quad (7.4)$$

The two signs are thus identified by a discrete isometry of the S1 family. Because the finite observable records only the abstract causal order,

$$\boxed{\mu_{N,\varepsilon}^{[P]}(C) = \mu_{N,-\varepsilon}^{[P]}(C)} \quad (N \geq 2, C \in \mathcal{C}_N). \quad (7.5)$$

Theorem 5 (exact parity of the antisymmetric orbit). *For the path γ_ψ in (7.3), every finite unlabeled-poset law is a real-analytic even function of ε . Consequently all its odd jets at the reference point vanish, in particular*

$$\left. \frac{d}{d\varepsilon} \mu_{N,\varepsilon}^{[P]} \right|_{\varepsilon=0} = 0 \quad \forall N \geq 2. \quad (7.6)$$

Proof. If $p_\pi(\varepsilon)$ is the probability of the rank permutation $\pi \in S_N$, then

$$p_\pi(\varepsilon) = \frac{\langle e^{2\varepsilon T_\pi} \rangle_0}{N! Z(\varepsilon)^N}, \quad T_\pi := \sum_{i=1}^N \psi(U_{(i)}, V_{(\pi(i))}), \quad (7.7)$$

where the bracket is expectation over two independent vectors of uniform order statistics. Boundedness of ψ permits differentiation under the integral to every order, so p_π is real analytic. Coordinate exchange gives $p_\pi(-\varepsilon) = p_{\pi^{-1}}(\varepsilon)$. Each fiber Γ_C is closed under inversion (§3); summing (7.7) over the fiber proves (7.5), hence the assertion. $\square$

To measure the first nonvanishing response of the complete finite law, define the path invariant

$$r_N(\gamma_\psi) := \inf \left\{ k \geq 1 : \left. \frac{d^k}{d\varepsilon^k} \mu_{N,\varepsilon}^{[P]} \right|_{\varepsilon=0} \neq 0 \text{ as a vector on } \mathcal{C}_N \right\}, \quad (7.8)$$

with $r_N(\gamma_\psi) = \infty$ if every jet vanishes. The notation refers to the full path, not merely to its zero first-order tangent.

The same finite-likelihood formula gives

$$p'_\pi(0) = \frac{2}{N!} \langle T_\pi \rangle_0, \quad p''_\pi(0) = \frac{4}{N!} \left(\langle T_\pi^2 \rangle_0 - N \|\psi\|_{L^2(D)}^2 \right). \quad (7.9)$$

At $N = 2$, exact polynomial integration over the two order simplices yields

$$\boxed{\left. \frac{d^2}{d\varepsilon^2} \mu_{2,\varepsilon}^{[P]}(\text{antichain}) \right|_0 = \frac{8}{5}, \quad \left. \frac{d^2}{d\varepsilon^2} \mu_{2,\varepsilon}^{[P]}(\text{chain}) \right|_0 = -\frac{8}{5}.} \quad (7.10)$$

Equivalently,

$$\mu_{2,\varepsilon}^{[P]}(\text{antichain}) = \frac{1}{2} + \frac{4}{5}\varepsilon^2 + O(\varepsilon^4), \quad \mu_{2,\varepsilon}^{[P]}(\text{chain}) = \frac{1}{2} - \frac{4}{5}\varepsilon^2 + O(\varepsilon^4). \quad (7.11)$$

Thus the magnitude of the deformation is locally visible already at the smallest cardinality supporting two distinct causal orders, while its sign remains identified by (7.4).

For $m \geq 3$, define the parameter-independent deletion kernel

$$K_{m,m-1}(C, D) := \frac{1}{m} \# \{v \in C : [C \setminus \{v\}] = D\}, \quad (7.12)$$

and let

$$K_{N \rightarrow 2} := K_{3,2} \circ K_{4,3} \circ \cdots \circ K_{N,N-1}, \quad K_{2 \rightarrow 2} := I. \quad (7.13)$$

Uniformly deleting points from an iid sample leaves an iid subsample, so for every ε ,

$$\mu_{2,\varepsilon}^{[P]} = K_{N \rightarrow 2} \mu_{N,\varepsilon}^{[P]}. \quad (7.14)$$

Since the kernel does not depend on ε , differentiation gives

$$\left(\mu_2^{[P]}\right)^{(k)}(0) = K_{N \rightarrow 2} \left(\mu_N^{[P]}\right)^{(k)}(0) \quad (k \geq 1). \quad (7.15)$$

Corollary 6 (first nonzero jet at every cardinality). *For the explicit path (7.3),*

$$\boxed{r_N(\gamma_\psi) = 2 \quad \forall N \geq 2.} \quad (7.16)$$

Proof. Theorem 5 gives $r_N(\gamma_\psi) \geq 2$. If $(\mu_N^{[P]})''(0)$ were zero, (7.15) with $k = 2$ would force $(\mu_2^{[P]})''(0) = 0$, contradicting (7.10). Hence the second jet is nonzero for every $N \geq 2$, proving (7.16). $\square$

Corollary 6 is an existence statement for one admissible orbit. It neither classifies the second differential on $\bigwedge^2 H$ nor asserts that every antisymmetric direction has order two. No general operator Q_N , quadratic null cone, estimator, rate, or nonlinear reconstruction is introduced. The first-order zero is explained by the exact isometric fold $\varepsilon \leftrightarrow -\varepsilon$; it must not be described as physical information loss.

Appendix G gives the finite-likelihood derivatives, the exact $N = 2$ integration, and the deletion-kernel argument.

8 Relation to prior work

Bombelli (2000) already sets up the framework studied here: the complete law of an unlabeled causal poset at fixed cardinality, and a statistical comparison between two such laws obtained from different geometries. Janson (2011) supplies the general limiting framework of poset kernels and consistent finite laws in which this construction sits. Surya (2026) tells a related story: increasing resolution can lift degeneracies of a causal compression, though through a different observable — expected interval abundances rather than the full unlabeled-poset law used here. None of the three computes the differential of the finite- N law at a reference geometry, its rank, or its kernel. That computation is what §§2–7 do.

The many-to-one correspondence between permutations and two-dimensional posets that this differential must be summed over is itself classical. Bayoumi et al. (1994) work explicitly with this correspondence, its realizers, the closure of a fiber under $\sigma \mapsto \sigma^{-1}$, and the near-uniqueness of realizers for prime posets. Recovering that correspondence is not the contribution here; we instead sum a differential score representative over the entire fiber,

$$A_C = \sum_{\sigma \in \Gamma_C} P_\sigma,$$

and ask what these class-sum representatives span once we pass to unlabeled poset classes.

Before that quotient is taken, the relevant differential structure is close to two existing constructions. Even-Zohar (2020) decomposes the space of pattern densities via the representation theory of S_N and isolates the standard-representation block of dimension $(N - 1)^2$, realized through permutation matrices compressed to $\mathbf{1}^\perp$; his asymptotic regime concerns fluctuations of a random permutation’s pattern profile as the host size grows, a scaling question distinct from the local ε -derivative used here.

The comparison with Kurečka (2022) can be made exact using (1.1). For $t = (t_\pi)_{\pi \in S_N}$, put $M(t) = \sum_\pi t_\pi P_\pi$. Kurečka’s Lemma 9 expresses every coefficient of the gradient polynomial as a nonzero scalar multiple of $b_i^T M(t) b_j$, where $b_2, \dots, b_N$ form a basis of E_N . Hence

$$\ker(\text{gradient-polynomial map}) = \ker T_N. \quad (8.1)$$

Because $M(t)$ has every row and column sum equal to $\sum_{\pi} t_{\pi}$, the condition $T_N(t) = 0$ is equivalent to $M(t)$ being constant, which is Kurečka’s Lemma 12. Thus the permutation-level differential, the Bernstein basis, the compression to E_N , the covering-matrix technique, and the ambient kernel in (8.1) all belong to that prior work.

The causal-order quotient adds the first arrow of (1.1). Knowing the ambient kernel rewrites, but does not solve, the restricted-image problem:

$$\text{rank}(T_N J_N) = \dim(\text{im } J_N) - \dim(\text{im } J_N \cap \ker T_N). \quad (8.2)$$

Kurečka does not study the fiber-constant subspace $\text{im } J_N$ or determine the intersection in (8.2). Inversion-closure of every Γ_C gives only $T_N(\text{im } J_N) \subseteq \text{Sym}(E_N)$. The near-chain construction of §4 supplies the reverse inclusion and proves the exact residual statement

$$\boxed{T_N(\text{im } J_N) = \text{Sym}(E_N)}. \quad (8.3)$$

This is the class-sum span assertion of Theorem 1. The statement is narrower than a claim of novelty for the ambient differential or for the abstract symmetric module. Chan et al. (2020) and Garbe et al. (2025) are adjacent permuton-forcing and feasible-region results, on quasirandomness-forcing pattern sums and on the dimension of the feasible region of pattern densities respectively, and they do not give this fiber-indexed span statement either.

The abstract target module and its dimension already appear in the representation-theory literature on rankings. Diaconis (1989) decomposes functions on rankings via S_N representation theory and gives, for unordered-pair effects, $M^{(N-2,2)} \simeq S^{(N)} \oplus S^{(N-1,1)} \oplus S^{(N-2,2)}$, already the module, and the dimension $\binom{N}{2}$, behind a Johnson-scheme reformulation of $\text{Sym}(E_N)$. The monograph (Diaconis, 1988) makes this concrete with a worked instance on unordered pairs of experimental varieties, the Diallel Cross Design, and develops the associated model family. Neither work introduces unlabeled two-dimensional-poset fibers, their class sums, or a rank theorem for those specific sums. So $\text{Sym}(E_N)$ and its dimension are not a new representation here; the narrower claim we are actually making is that sums over isomorphism fibers of unlabeled two-dimensional posets generate exactly that module.

Fisher information after passing from continuous observations to ranks is not a new idea either. Hallin et al. (2001) already have Bernstein-type polynomials showing up in Hájek projections of rank statistics, though asymptotically rather than at our exact finite N . Hoff (2007) establishes the rank likelihood as a marginal-free semiparametric likelihood, and Hoff et al. (2014) together with Sei and Matsumoto (2020) develop the induced information and divergence of Gaussian-copula and rank models, including finite-sample identifiability loss. None of these reaches the further quotient from a full rank permutation to an unlabeled poset, $\Pi_N \rightarrow [P_{\Pi_N}]$, or classifies its S1 support the way §§3–6 do.

The operator identity connecting these two levels is standard once its kernel is known. Pollard (2013) shows, in the differentiability-in-quadratic-mean framework, that the score of a statistic is the conditional expectation of the original score; on a Hilbert space, any bounded linear operator factors tautologically through the orthogonal projection onto the complement of its kernel, with an injective restriction to that complement. So the factorization $D\mathcal{S}_N = B_N P_N^{\text{vis}}$, used from §5 on, is not an independent operator construction. Theorem 1 is what pins the complement down exactly, $(\ker D\mathcal{S}_N)^{\perp} = \text{Sym}^2 P_{N-1}$, and the injectivity of B_N follows immediately from that.

§7 sits against several established mechanisms rather than introducing them. Rotnitzky et al. (2000) study likelihood models with singular information, relating the order of the first nonvanishing derivative to inferential behavior and noting the sign ambiguity that arises when that order is even; a law invisible at first order and visible only at a higher one, and the role parity plays in that, is not

a general contribution of this paper. Within the permuton literature, forcing results go beyond the gradient: Chan (2021) and Crudele et al. (2024) compute Hessians of pattern-density combinations around the uniform permuton once the gradient vanishes, precedent for using a second differential of a permutation-pattern law to detect perturbations hidden at first order. Projective consistency under uniform deletion is also standard. Theorem 5 and Corollary 6 combine these ingredients for one explicit antisymmetric S1 orbit; they do not introduce a general higher-order classification.

Put together, the precedents above cover the permutation-level differential, the abstract representation-theoretic target, and the general mechanics of statistic-induced scores and singular first-order information. This paper determines what is left over: the exact effect of the additional quotient from labeled rank permutations to unlabeled finite causal-order laws, at every fixed cardinality N .

9 Discussion: causal compression

The phrase *causal compression* is a compact name for the exact S1 factorization

$$D\mathcal{S}_N = B_N P_N^{\text{vis}}, \quad V_N = \text{Sym}^2 P_{N-1}. \quad (9.1)$$

Read literally, at the level actually proved here: the first differential of an infinite-dimensional interaction tangent is compressed to the finite-dimensional component $P_N^{\text{vis}} f$, and that component is then encoded injectively as a score of the finite unlabeled-poset law. The projection selects what is visible, B_N performs the statistical encoding, and $F_N = B_N^* B_N$ measures its directional strength. These are three different parts of one channel, not three names for the same operator.

The word *causal* here is not decorative. In the $1 + 1$ -dimensional diamond, the order relation is fixed by the joint pattern of the two null rankings. Passing from continuous coordinates to ranks throws away marginal information; passing from a rank permutation to an unlabeled poset throws away the choice of linear realizer. What survives both steps at first order is exactly the symmetric polynomial interaction sector in (9.1). In this restricted, model-specific sense the causal past and future compress the present: the finite order does not preserve a perturbation point by point, it records a finite collection of symmetric interaction modes through the relations among the sampled events.

Resolution grows, but not uniformly. Theorem 1 and Corollary 2 turn this interpretation into an exact filtration,

$$V_N \subsetneq V_{N+1}, \quad \dim V_N = \binom{N}{2}, \quad \overline{\bigcup_{N \geq 2} V_N} = H \widehat{\otimes}_{\text{sym}} H. \quad (9.2)$$

Each additional cardinality opens genuinely new polynomial directions, so no fixed nonzero symmetric tangent remains invisible at every sufficiently large resolution. This statement is nevertheless pointwise in the tangent. The moving direction $p_N \otimes p_N$ lies beyond the resolution of the size- N experiment, and Theorem 4 correspondingly gives

$$\widehat{F}_N \xrightarrow{\text{SOT}} \Pi_{\text{sym}}, \quad \|\widehat{F}_N - \Pi_{\text{sym}}\| \geq 1. \quad (9.3)$$

Thus increasing N resolves every fixed symmetric interaction in the limit, but there is no uniform resolution over the Hilbert–Schmidt unit sphere and no operator-norm convergence.

Visibility is not sensitivity. The exact spectra at $N = 2, 3, 4$ show that two directions can both belong to V_N and nevertheless be encoded with very different Fisher strengths. At $N = 4$, visible polynomial modes even mix before the Fisher eigenvectors emerge. Hence the binary question

answered by Theorem 1 — whether a direction survives the differential — is logically prior to, but does not answer, the quantitative question treated by Theorem 4. In particular, neither F_N nor its normalized ambient extension is a finite- N support projection.

A first-order kernel is not a nonlinear erasure theorem. The full antisymmetric sector belongs to $\ker D\mathcal{S}_N$ at every cardinality, but §7 shows why that fact must not be called physical information loss. For the explicit admissible odd path γ_ψ , coordinate exchange is an isometry satisfying $\iota^*g_\varepsilon = g_{-\varepsilon}$; the finite law therefore folds the two signs together. Its linear term vanishes by symmetry, while

$$r_N(\gamma_\psi) = 2 \quad \forall N \geq 2. \quad (9.4)$$

The magnitude of this deformation is detected at second order at every nontrivial finite resolution. This single orbit proves that membership in the first-order kernel does not imply invariance of the full nonlinear law; it does not classify higher-order behavior throughout $\bigwedge^2 H$.

§§3–7 together describe the local anatomy of one channel. Score representatives give its finite linear functionals; the all- N span theorem pins down their exact support; the quotient marks off the equivalence class the differential identifies; the Fisher operator then resolves directions within that support, and the antisymmetric orbit of §7 shows that a symmetry-forced first-order zero can still carry a nonzero second jet. The result is a statement about tangent visibility and statistical resolution for the S1 finite-law experiment, not a reconstruction theorem.

In particular, *the causal past and future compress the present* must not be read as saying that causal data universally determine a present geometry. The results do not reconstruct continuous coordinates or a Lorentzian metric from one causet, establish global injectivity, realize every Hilbert–Schmidt tangent geometrically, or extend beyond S1 and $1+1$ dimensions. The phrase names the exact model-specific compression in (9.1), and nothing broader.

10 Limitations and open problems

Model scope. Every theorem in this paper concerns the explicit S1 interaction model in a $(1+1)$ -dimensional causal diamond, expanded at the independent reference point. The arguments do not establish an analogue in $(2+1)$ or $(3+1)$ dimensions, for a general Lorentzian spacetime, or for an arbitrary causal-set sampling model. Such extensions would require new geometric and combinatorial input and are outside the present paper.

Ambient tangents versus geometric realizability. The Hilbert space $\mathcal{X} = H \hat{\otimes} H$ is the analytic domain of the score operators. The exact identities for V_N , $\ker D\mathcal{S}_N$, and F_N therefore classify the finite channel on that ambient tangent space. They do not prove that every Hilbert–Schmidt direction is generated by an admissible curve of Lorentzian geometries. Geometric realizability of an arbitrary ambient tangent remains open and is not needed for the finite-channel classification.

Differential identification versus nonlinear reconstruction. Corollary 3 identifies exactly the quotient seen by $D\mathcal{S}_N$; it does not imply injectivity of the full map $\mathcal{S}_N$ at finite distance, recovery of coordinates or a metric from one causet, or reconstruction from the family of finite laws. The antisymmetric orbit of §7 makes the distinction concrete: its first jet vanishes and its second jet does not. That calculation treats one explicit admissible orbit only. A general second-order operator Q_N , the associated quadratic null cone, and a classification of the antisymmetric sector are not developed here.

Pointwise asymptotic retention. The convergence $\hat{F}_N \rightarrow \Pi_{\text{sym}}$ is strong-operator convergence. It gives asymptotic retention for each fixed symmetric Hilbert–Schmidt tangent, but no uniform rate over the unit sphere. Indeed, the moving unresolved directions in (6.26) give $\|\hat{F}_N - \Pi_{\text{sym}}\| \geq 1$ for

every N , and hence rule out operator-norm convergence. The exact Fisher spectra computed here are confined to $N = 2, 3, 4$; no all- N spectral formula, uniform conditioning bound, or nonlinear estimator is claimed.

Priority status. The statistical comparison of finite causal-order laws, and the general expectation that larger samples can sharpen resolution, both have clear precedents; §8 reviews them, together with substantial partial precedents for the specific ingredients behind our results. We have not found an exact counterpart of the S1 all- N class-sum span theorem, or of the antisymmetric-orbit statement, in the literature considered there, but that absence is not itself a priority claim, and our search was not exhaustive. A broader specialist review would be needed before any affirmative claim of novelty.

11 Conclusion

The finite unlabeled causal-order law has, at the independent reference point of the $1 + 1$ -dimensional S1 model, a local differential structure we can now state exactly. The score representatives of §3 reduce the problem to class sums over the fibers of the permutation-to-poset map, and the constructive all- N argument of §4 gives

$$\text{span}\{R_C^{(N)} : C \in \mathcal{C}_N\} = V_N = \text{Sym}^2 P_{N-1}, \quad \dim V_N = \binom{N}{2} \quad (N \geq 2). \quad (11.1)$$

These are exactly the first-order interaction modes that survive the additional quotient from a rank permutation to an unlabeled causal order.

The operator picture makes this precise. For every N ,

$$D\mathcal{S}_N = B_N P_N^{\text{vis}}, \quad \ker D\mathcal{S}_N = V_N^{\perp \text{sym}} \oplus \bigwedge^2 H, \quad \mathcal{X} / \ker D\mathcal{S}_N \simeq V_N. \quad (11.2)$$

The visible spaces are strictly nested, and their union is dense in the symmetric interaction Hilbert space: increasing cardinality keeps opening new first-order directions, while (11.2) itself states only differential identifiability at the reference model.

Visibility is one thing, statistical sensitivity another. The positive operator $F_N = B_N^* B_N$ resolves directions inside V_N , and the exact spectra at $N = 2, 3, 4$ show both strong anisotropy and, at $N = 4$, modal mixing. Normalized by the Fisher information of N continuous copula observations, Theorem 4 gives

$$\widehat{F}_N \xrightarrow{\text{SOT}} \Pi_{\text{sym}}, \quad \|\widehat{F}_N - \Pi_{\text{sym}}\| \geq 1 \quad \text{for every } N. \quad (11.3)$$

Every fixed symmetric Hilbert–Schmidt tangent is asymptotically retained, but not uniformly over the unit sphere; keeping the two stages, continuous observations $\rightarrow$ ranks and ranks $\rightarrow$ unlabeled posets, separate is what lets us locate the corresponding Fisher losses.

The explicit antisymmetric orbit of §7 completes the picture. Coordinate exchange folds g_ε onto the isometric geometry $g_{-\varepsilon}$, so the finite laws are even and their first derivatives vanish. Corollary 6 combines the exact $N = 2$ second derivative with projective consistency under uniform deletion to give one admissible second-order witness at every resolution, not a classification of the antisymmetric sector.

Three layers make up the resulting picture:

tangent visibility = V_N , statistical resolution = F_N , higher-order detectability = $r_N(\gamma_\psi) = 2$ for the explicit witness.	(11.4)
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This is an exact, local account of what the finite causal-order channel retains in S1, not a reconstruction of geometry from a causet, a claim of global or nonlinear identifiability, or a result beyond the 1 + 1-dimensional S1 model; §10 states the full limits. The statistical framework itself is inherited from prior work — what is isolated here is the explicit differential classification and its consequences within that fixed scope.

A QMD and score representatives

Fix an admissible generator $\psi \in C([0, 1]^2)$, put $f = \mathcal{P}\psi$, and let c_ε be the copula density obtained from the normalized S1 density by the two marginal probability integral transforms of §2. The calculation there gives, uniformly on the unit square,

$$c_\varepsilon(u, v) = 1 + 2\varepsilon f(u, v) + o(\varepsilon), \quad \int_0^1 f(u, v) du = \int_0^1 f(u, v) dv = 0. \quad (\text{A.1})$$

Positivity of the exponential S1 density and continuity on the compact domain give a common positive lower bound for c_ε when $|\varepsilon|$ is small. Taylor expansion of the square root, with the uniform remainder in (A.1), therefore yields

$$\int_{[0,1]^2} (\sqrt{c_\varepsilon} - 1 - \varepsilon f)^2 du dv = o(\varepsilon^2). \quad (\text{A.2})$$

So the one-observation copula experiment is QMD at zero with score $2f = h_\psi$.

For a fixed N , write $L_{N,\varepsilon} = \prod_{k=1}^N c_\varepsilon(U_k, V_k)$ for the density of the iid copula sample with respect to Lebesgue measure on $([0, 1]^2)^N$. Taking the finite product in (A.2) gives

$$\int \left(\sqrt{L_{N,\varepsilon}} - 1 - \varepsilon \sum_{k=1}^N f(U_k, V_k) \right)^2 = o(\varepsilon^2). \quad (\text{A.3})$$

Hence the full-sample score is

$$T_{N,\psi} = 2 \sum_{k=1}^N f(U_k, V_k) = \sum_{k=1}^N h_\psi(U_k, V_k). \quad (\text{A.4})$$

Let $\mathcal{A}_\sigma$ be the set of samples whose relative rank permutation is $\sigma \in S_N$. This event is defined entirely by strict coordinate inequalities and is independent of ε ; ties have probability zero. Since c_ε and its parameter derivative are uniformly bounded in a neighborhood of zero, differentiation under the integral over $\mathcal{A}_\sigma$ is valid. With $p_\varepsilon(\sigma) = \int_{\mathcal{A}_\sigma} L_{N,\varepsilon}$ and $p_0(\sigma) = 1/N!$,

$$\begin{aligned} p'_\sigma(0; f) &= \mathbb{E}_0[\mathbf{1}_{\{\Pi_N = \sigma\}} T_{N,\psi}], \\ S_N^\Pi(f)(\sigma) &:= \partial_\varepsilon \log p_\varepsilon(\sigma)|_0 = \mathbb{E}_0[T_{N,\psi} \mid \Pi_N = \sigma]. \end{aligned} \quad (\text{A.5})$$

This is the conditional-score identity used in §2; it follows from the likelihood and does not assume that the ordered observations remain independent after conditioning.

Under the uniform reference law, the two vectors of order statistics are independent, and conditional on $\Pi_N = \sigma$ the point of U -rank i is paired with the point of V -rank $\sigma(i)$. If $d_i^{(N)}$ is the order-statistic density from (3.1), (A.4)–(A.5) give

$$S_N^\Pi(f)(\sigma) = 2 \sum_{i=1}^N \left\langle f, d_i^{(N)} \otimes d_{\sigma(i)}^{(N)} \right\rangle, \quad (\text{A.6})$$

and consequently

$$p'_\sigma(0; f) = \frac{2}{N!} \sum_{i=1}^N \langle f, d_i^{(N)} \otimes d_{\sigma(i)}^{(N)} \rangle. \quad (\text{A.7})$$

Because both marginals of f vanish, writing $d_i^{(N)} = 1 + b_i^{(N)}$ removes the constant and one-coordinate terms:

$$\langle f, d_i^{(N)} \otimes d_j^{(N)} \rangle = \langle f, b_i^{(N)} \otimes b_j^{(N)} \rangle. \quad (\text{A.8})$$

Combining (A.7)–(A.8) proves the representative formula

$$R_\sigma^{(N)} = \frac{2}{N!} \sum_{i=1}^N b_i^{(N)} \otimes b_{\sigma(i)}^{(N)}, \quad p'_\sigma(0; f) = \langle f, R_\sigma^{(N)} \rangle. \quad (\text{A.9})$$

The last step is to pass from labeled ranks to the observable unlabeled poset. For $C \in \mathcal{C}_N$, the fiber Γ_C is finite and fixed, so direct summation of (A.9) gives

$$\begin{aligned} \mu_{N,0}^{[P]}(C) &= \frac{|\Gamma_C|}{N!}, \\ \partial_\varepsilon \mu_{N,\varepsilon}^{[P]}(C) \Big|_0 &= \langle f, R_C^{(N)} \rangle, \quad R_C^{(N)} := \sum_{\sigma \in \Gamma_C} R_\sigma^{(N)}. \end{aligned} \quad (\text{A.10})$$

Every reference mass in (A.10) is positive. Since $\mathcal{C}_N$ is finite, coordinatewise differentiability of its probabilities is equivalent here to the discrete QMD expansion

$$\sum_{C \in \mathcal{C}_N} \left[\sqrt{\mu_{N,\varepsilon}^{[P]}(C)} - \sqrt{\mu_{N,0}^{[P]}(C)} - \frac{\varepsilon}{2} (D\mathcal{S}_N f)(C) \sqrt{\mu_{N,0}^{[P]}(C)} \right]^2 = o(\varepsilon^2), \quad (\text{A.11})$$

with score

$$(D\mathcal{S}_N f)(C) = \frac{\langle f, R_C^{(N)} \rangle}{\mu_{N,0}^{[P]}(C)}. \quad (\text{A.12})$$

Also, $\sum_{C \in \mathcal{C}_N} R_C^{(N)} = 0$: after summing (A.9) over σ , each $b_j^{(N)}$ occurs $(N-1)!$ times at every fixed position and $\sum_j b_j^{(N)} = 0$. So (A.12) has zero mean, as a score must; polarization gives

$$G_{[P]}^{(N)}(f, g) = \sum_{C \in \mathcal{C}_N} \frac{\langle f, R_C^{(N)} \rangle \langle g, R_C^{(N)} \rangle}{\mu_{N,0}^{[P]}(C)}. \quad (\text{A.13})$$

Equations (A.9)–(A.13) were derived for continuous S1 tangents. Since every $R_C^{(N)}$ is a finite sum of polynomial tensors, the right-hand sides define bounded linear functionals on $\mathcal{X} = H \widehat{\otimes} H$. This is the Hilbert-space extension used in §§3–6; it is not a claim that every element of $\mathcal{X}$ is geometrically realizable.

B Finite reduction to $\text{Sym}(E_N)$

Let

$$E_N = \mathbf{1}^\perp \subset \mathbb{R}^N, \quad P_{N-1} = \text{span}\{p_1, \dots, p_{N-1}\} \subset H. \quad (\text{B.1})$$

The functions $d_i^{(N)}/N$ form the Bernstein basis of the polynomials of degree at most $N - 1$. Since $1 = N^{-1} \sum_i d_i^{(N)}$, their centered versions satisfy

$$b_i^{(N)} = d_i^{(N)} - \frac{1}{N} \sum_{j=1}^N d_j^{(N)}, \quad \sum_{i=1}^N b_i^{(N)} = 0. \quad (\text{B.2})$$

This is their only linear relation. Indeed, every coefficient vector $a \in \mathbb{R}^N$ decomposes as $a = \bar{a}\mathbf{1} + z$ with $z \in E_N$; the constant part gives the relation in (B.2), whereas

$$\sum_{i=1}^N z_i b_i^{(N)} = \sum_{i=1}^N z_i d_i^{(N)}, \quad (\text{B.3})$$

and linear independence of the Bernstein basis makes the right-hand side zero only when $z = 0$. Consequently,

$$\Lambda_N : E_N \longrightarrow P_{N-1}, \quad \Lambda_N z = \sum_{i=1}^N z_i b_i^{(N)}, \quad (\text{B.4})$$

is injective and, since both spaces have dimension $N - 1$, is an isomorphism.

We use the Euclidean inner product to identify $\text{Sym}(E_N)$, the self-adjoint endomorphisms of E_N , with $\text{Sym}^2 E_N$. For $M \in \text{Sym}(E_N)$, let $\widetilde{M}$ be its self-adjoint extension to $\mathbb{R}^N$ that vanishes on $\text{span}\{\mathbf{1}\}$. The tensor transport induced by (B.4) is

$$\begin{aligned} \mathfrak{T}_N : \text{Sym}(E_N) &\longrightarrow \text{Sym}^2 P_{N-1}, \\ M &\longmapsto \sum_{i,j=1}^N \widetilde{M}_{ij} b_i^{(N)} \otimes b_j^{(N)}, \end{aligned} \quad (\text{B.5})$$

Equivalently, $\mathfrak{T}_N$ is $\Lambda_N \otimes \Lambda_N$ restricted to the symmetric tensor square. It is therefore a linear isomorphism. It need not be an isometry; only spans and ranks, not Fisher eigenvalues, are preserved by this reduction.

With the convention of (3.12),

$$P_\sigma = \sum_{i=1}^N e_i e_{\sigma(i)}^\top, \quad A_C = \sum_{\sigma \in \Gamma_C} P_\sigma. \quad (\text{B.6})$$

Every permutation matrix has unit row and column sums. Hence

$$A_C \mathbf{1} = |\Gamma_C| \mathbf{1}, \quad \mathbf{1}^\top A_C = |\Gamma_C| \mathbf{1}^\top, \quad (\text{B.7})$$

so E_N is invariant under A_C . Also, $P_\sigma^\top = P_{\sigma^{-1}}$, and the fiber Γ_C is closed under inversion (§3). So

$$A_C^\top = A_C, \quad A_C|_{E_N} \in \text{Sym}(E_N). \quad (\text{B.8})$$

Because $\sum_i b_i^{(N)} = 0$, projecting onto E_N on either matrix index does not change the transported tensor. Equations (3.13) and (B.5) therefore give the exact identity

$$R_C^{(N)} = \frac{2}{N!} \mathfrak{T}_N(A_C|_{E_N}). \quad (\text{B.9})$$

The scalar $2/N!$ is nonzero for every N . Since $\mathfrak{T}_N$ is an isomorphism, (B.9) proves

$$\text{span}\{R_C^{(N)} : C \in \mathcal{C}_N\} = \text{Sym}^2 P_{N-1} \iff \text{span}\{A_C|_{E_N} : C \in \mathcal{C}_N\} = \text{Sym}(E_N). \quad (\text{B.10})$$

In particular, after choosing any basis of E_N , the right-hand side is equivalent to the finite rank condition

$$\text{rank}(\text{vec}_{\text{sym}}(A_C|_{E_N}))_{C \in \mathcal{C}_N} = \frac{N(N-1)}{2}. \quad (\text{B.11})$$

Appendix C proves this rank structurally from the almost-chain family; no finite-cardinality enumeration is part of the reduction above.

C Almost-chain classes and Laplacian triangularization

Fix $N \geq 2$. For each pair of integers $0 \leq a < b \leq N-1$, let $C_{a,b}$ be the poset consisting of a chain

$$c_1 < \cdots < c_{N-1} \quad (\text{C.1})$$

and one further element z , with

$$c_i < z \quad (i \leq a), \quad z < c_i \quad (i > b), \quad z \parallel c_i \quad (a < i \leq b). \quad (\text{C.2})$$

Every linear extension of $C_{a,b}$ is obtained by inserting z after exactly $k \in \{a, \dots, b\}$ elements of the chain; denote this extension by L_k . For two such extensions, their intersection places $c_1, \dots, c_{\min(s,t)}$ below z and $c_{\max(s,t)+1}, \dots, c_{N-1}$ above it. Therefore

$$L_s \cap L_t = C_{a,b} \iff \{s, t\} = \{a, b\}. \quad (\text{C.3})$$

Any σ with $P_\sigma \cong C_{a,b}$ pulls the natural order and the σ -order back to an ordered realizer pair of $C_{a,b}$. Conversely, enumerating the elements in the first order of any ordered realizer pair and recording their ranks in the second produces one such σ . Applying an automorphism or a simultaneous relabeling to both orders does not change this relative-rank permutation, so no additional permutations arise from the choice of an isomorphism.

So the only ordered realizer pairs are (L_a, L_b) and (L_b, L_a) . After normalizing the first extension to the natural order, the relative permutation is a cycle $\tau_{a,b}$ on the consecutive interval

$$I_{a,b} = \{a+1, a+2, \dots, b+1\}, \quad (\text{C.4})$$

and reversing the ordered pair gives its inverse. Hence

$$\Gamma_{C_{a,b}} = \{\tau_{a,b}, \tau_{a,b}^{-1}\}, \quad (\text{C.5})$$

as a set without multiplicity. When $b = a+1$, the cycle is a transposition and the two displayed permutations coincide.

These $\binom{N}{2}$ classes are pairwise distinct. Indeed, the multiset of strict-past cardinalities is

$$\{|\text{Past}(y)| : y \in C_{a,b}\} = \{0, 1, \dots, b-1, b+1, \dots, N-1\} \uplus \{a\}. \quad (\text{C.6})$$

It omits b and contains a twice, so it determines (a, b) . This means the construction supplies exactly one distinct class for each of the $\binom{N}{2}$ pairs $a < b$.

For $1 \leq i < j \leq N$, define the edge Laplacian

$$L_{ij} := (e_i - e_j)(e_i - e_j)^\top. \quad (\text{C.7})$$

Each L_{ij} annihilates $\mathbf{1}$ and preserves E_N . If a linear combination of their restrictions vanishes on E_N , it also vanishes on $\text{span}\{\mathbf{1}\}$, hence on all of $\mathbb{R}^N$; its (i, j) entry is $-w_{ij}$, so every coefficient is zero. There are $\binom{N}{2} = \dim \text{Sym}(E_N)$ such matrices, and therefore

$$\{L_{ij}|_{E_N} : 1 \leq i < j \leq N\} \text{ is a basis of } \text{Sym}(E_N). \quad (\text{C.8})$$

Summing all edges gives the complete-graph Laplacian

$$\sum_{1 \leq i < j \leq N} L_{ij} = NI - \mathbf{1}\mathbf{1}^\top, \quad \sum_{i < j} L_{ij}|_{E_N} = NI_{E_N}. \quad (\text{C.9})$$

Symmetrize the interval cycle by setting

$$S_{a,b} := P_{\tau_{a,b}} + P_{\tau_{a,b}}^\top. \quad (\text{C.10})$$

Equation (C.5) implies

$$S_{a,b} = \begin{cases} 2A_{C_{a,b}}, & b = a + 1, \\ A_{C_{a,b}}, & b > a + 1. \end{cases} \quad (\text{C.11})$$

The scalar relating $S_{a,b}$ to the class sum is nonzero in both cases. On E_N , put

$$Q_{a,b} := 2I_{E_N} - S_{a,b}|_{E_N}. \quad (\text{C.12})$$

The matrix $Q_{a,b}$ is the graph Laplacian of the consecutive cycle on $I_{a,b}$, with the unique edge counted twice when the interval has length two. In the formulas below, each L_{ij} is understood as its restriction to E_N . Thus

$$Q_{a,a+1} = 2L_{a+1,a+2}, \quad (\text{C.13})$$

whereas, when $b > a + 1$,

$$Q_{a,b} = L_{a+1,b+1} + \sum_{k=a+1}^b L_{k,k+1}. \quad (\text{C.14})$$

These identities are triangular in the interval length. They invert as

$$L_{i,i+1} = \frac{1}{2}Q_{i-1,i}, \quad L_{ij} = Q_{i-1,j-1} - \frac{1}{2} \sum_{k=i}^{j-1} Q_{k-1,k} \quad (j > i + 1). \quad (\text{C.15})$$

By (C.8),

$$\text{span}\{Q_{a,b} : 0 \leq a < b \leq N - 1\} = \text{Sym}(E_N). \quad (\text{C.16})$$

The common identity term in (C.12) still has to be removed. From (C.9) and (C.15), there are coefficients $c_{a,b}$ such that

$$I_{E_N} = \sum_{a < b} c_{a,b} Q_{a,b}. \quad (\text{C.17})$$

Their individual values are unnecessary, but their sum is not. An edge L_{ij} at distance $d = j - i$ contributes total coefficient $1 - d/2$ in its expression through the Q 's: this is $1/2$ for $d = 1$, and for $d > 1$ it is one long-interval term minus d adjacent terms of coefficient $1/2$. Since there are $N - d$ edges at distance d , (C.9) gives

$$\begin{aligned} s_N := \sum_{a < b} c_{a,b} &= \frac{1}{N} \sum_{d=1}^{N-1} (N - d) \left(1 - \frac{d}{2}\right) \\ &= \frac{(N-1)(5-N)}{12}. \end{aligned} \quad (\text{C.18})$$

Substituting $Q_{a,b} = 2I_{E_N} - S_{a,b}|_{E_N}$ into (C.17) yields

$$(1 - 2s_N)I_{E_N} = - \sum_{a < b} c_{a,b} S_{a,b}|_{E_N}. \quad (\text{C.19})$$

The coefficient of the identity never vanishes:

$$1 - 2s_N = \frac{N^2 - 6N + 11}{6} = \frac{(N - 3)^2 + 2}{6} > 0. \quad (\text{C.20})$$

So I_{E_N} lies in the span of the $S_{a,b}|_{E_N}$. Equation (C.12) then puts every $Q_{a,b}$ in that same span, and (C.16) gives

$$\text{span}\{S_{a,b}|_{E_N} : a < b\} = \text{Sym}(E_N). \quad (\text{C.21})$$

The remaining step, (C.11), shows that the selected class sums themselves span $\text{Sym}(E_N)$. Appendix B then transports (C.21) to $V_N = \text{Sym}^2 P_{N-1}$, completing the constructive proof of Theorem 1 for every $N \geq 2$.

D Kernel, strict nesting, and density

Let

$$\mathcal{X} = H \widehat{\otimes} H, \quad \mathcal{X}_{\text{sym}} = H \widehat{\otimes}_{\text{sym}} H, \quad \mathcal{X}_{\text{alt}} = \bigwedge^2 H. \quad (\text{D.1})$$

The coordinate-swap involution $(\mathfrak{s}f)(u, v) = f(v, u)$ is unitary and self-adjoint. Its two eigenspaces are orthogonal, with projections

$$\Pi_{\text{sym}} = \frac{I + \mathfrak{s}}{2}, \quad \Pi_{\text{alt}} = \frac{I - \mathfrak{s}}{2}, \quad \mathcal{X} = \mathcal{X}_{\text{sym}} \oplus \mathcal{X}_{\text{alt}}. \quad (\text{D.2})$$

By (B.8)–(B.9), every class representative $R_C^{(N)}$ belongs to $\mathcal{X}_{\text{sym}}$. Appendix A gives

$$(D\mathcal{S}_N f)(C) = \frac{\langle f, R_C^{(N)} \rangle}{\mu_{N,0}^{[P]}(C)}, \quad \mu_{N,0}^{[P]}(C) > 0. \quad (\text{D.3})$$

Hence

$$\ker D\mathcal{S}_N = \ker G_{[P]}^{(N)} = \text{span}\{R_C^{(N)} : C \in \mathcal{C}_N\}^{\perp_{\mathcal{X}}}. \quad (\text{D.4})$$

The first equality uses $G_{[P]}^{(N)}(f, f) = \|D\mathcal{S}_N f\|_{\mathcal{K}_N}^2$. Theorem 1 identifies the span in (D.4) with $V_N = \text{Sym}^2 P_{N-1} \subset \mathcal{X}_{\text{sym}}$. Splitting the ambient orthogonal complement according to (D.2) proves

$$\ker D\mathcal{S}_N = V_N^{\perp_{\text{sym}}} \oplus \mathcal{X}_{\text{alt}} = (\text{Sym}^2 P_{N-1})^{\perp_{\text{sym}}} \oplus \bigwedge^2 H. \quad (\text{D.5})$$

Here $\perp_{\text{sym}}$ denotes the orthogonal complement inside $\mathcal{X}_{\text{sym}}$, not inside the full tensor product.

Let $P_N^{\text{vis}} = \Pi_{V_N} \Pi_{\text{sym}}$. Since $V_N \subset \mathcal{X}_{\text{sym}}$, this is the ambient orthogonal projection onto V_N , and (D.5) gives

$$\ker P_N^{\text{vis}} = \ker D\mathcal{S}_N. \quad (\text{D.6})$$

If $B_N = D\mathcal{S}_N|_{V_N}$, then B_N is injective: a vector in its kernel belongs simultaneously to V_N and $V_N^{\perp}$. For every $f \in \mathcal{X}$, (D.6) also gives

$$D\mathcal{S}_N f = D\mathcal{S}_N P_N^{\text{vis}} f = B_N P_N^{\text{vis}} f. \quad (\text{D.7})$$

This proves the exact factorization without identifying B_N with the support projection.

For completeness, let $q_N : \mathcal{X} \rightarrow \mathcal{X} / \ker D\mathcal{S}_N$ be the quotient map. The map

$$U_N : \mathcal{X} / \ker D\mathcal{S}_N \longrightarrow V_N, \quad U_N([f]) = P_N^{\text{vis}} f, \quad (\text{D.8})$$

is well defined by (D.6), is onto, and is injective for the same reason. The orthogonal decomposition $\mathcal{X} = V_N \oplus \ker D\mathcal{S}_N$ gives

$$\|[f]\|_{\mathcal{X} / \ker D\mathcal{S}_N} = \inf_{k \in \ker D\mathcal{S}_N} \|f + k\|_{\mathcal{X}} = \|P_N^{\text{vis}} f\|_{\mathcal{X}}. \quad (\text{D.9})$$

Thus U_N is a canonical isometric isomorphism and $D\mathcal{S}_N = B_N U_N q_N$. This is an identification of the differential quotient only.

We next prove strict nesting. Orthogonality of the shifted-Legendre basis gives

$$P_N = P_{N-1} \oplus \text{span}\{p_N\}. \quad (\text{D.10})$$

Write $x \odot y = x \otimes y + y \otimes x$. Taking symmetric tensor squares and using Theorem 1,

$$\begin{aligned} V_{N+1} &= \text{Sym}^2 P_N = \text{Sym}^2 P_{N-1} \oplus \{x \odot p_N : x \in P_{N-1}\} \oplus \text{span}\{p_N \otimes p_N\} \\ &= V_N \oplus \{x \odot p_N : x \in P_{N-1}\} \oplus \text{span}\{p_N \otimes p_N\}, \end{aligned} \quad (\text{D.11})$$

and the summands are orthogonal. In particular,

$$p_1 \odot p_N \in V_{N+1} \setminus V_N, \quad V_N \subsetneq V_{N+1} \quad (N \geq 2). \quad (\text{D.12})$$

The same witness gives the Fisher statement in Corollary 2. For the nonsymmetrized tangent $p_1 \otimes p_N$, its symmetric component is $(p_1 \odot p_N)/2$. Equation (D.5) annihilates it at size N , whereas injectivity of B_{N+1} on V_{N+1} gives

$$I_N^{[P]}(p_1 \otimes p_N) = 0, \quad I_{N+1}^{[P]}(p_1 \otimes p_N) > 0. \quad (\text{D.13})$$

Finally, centered polynomials are dense in $H = L_0^2([0, 1])$. Indeed, if ordinary polynomials q_m converge in L^2 to $h \in H$, then

$$\left\| q_m - \int_0^1 q_m - h \right\|_{L^2} \leq 2\|q_m - h\|_{L^2} \longrightarrow 0. \quad (\text{D.14})$$

Finite sums of elementary tensors from a dense subspace are dense in the Hilbert tensor product. Applying the continuous projection Π_{sym} , every symmetric Hilbert–Schmidt tensor can therefore be approximated by finite sums of symmetrized polynomial tensors. Each such finite sum belongs to $\text{Sym}^2 P_m = V_{m+1}$ for some m . So

$$\overline{\bigcup_{N \geq 2} V_N}^{\|\cdot\|_{\mathcal{X}}} = \mathcal{X}_{\text{sym}}. \quad (\text{D.15})$$

Combining (D.5), strict nesting, and (D.15) also gives

$$\bigcap_{N \geq 2} \ker D\mathcal{S}_N = \bigwedge^2 H. \quad (\text{D.16})$$

Equation (D.16) concerns simultaneous first-order invisibility across all finite resolutions. It does not say that antisymmetric geometric paths are invisible to the nonlinear law: the explicit path of §7 has a nonzero second jet. Likewise, the Hilbert-space density in (D.15) does not assert that every element of $\mathcal{X}_{\text{sym}}$ is generated by an admissible geometric curve.

E Hilbert–Schmidt bounds and Fisher retention

For $a \in H$, define the order-statistic transform

$$\mathcal{O}_N a := (\langle a, d_1^{(N)} \rangle, \dots, \langle a, d_N^{(N)} \rangle) \in \mathbb{R}^N, \quad (\text{E.1})$$

and, for $f \in \mathcal{X}$, define

$$H_{ij}^{(N)}(f) := \langle f, d_i^{(N)} \otimes d_j^{(N)} \rangle. \quad (\text{E.2})$$

So $H^{(N)}(f) = (\mathcal{O}_N \otimes \mathcal{O}_N)f$. Since $\sum_i d_i^{(N)} = N$ and both marginals of f vanish, this matrix has zero row and column sums. Appendix A gives the rank-permutation score as

$$S_N^\Pi(f)(\sigma) = 2 \sum_{i=1}^N H_{i,\sigma(i)}^{(N)}(f). \quad (\text{E.3})$$

If H and K have zero row and column sums and Π_N is uniform on S_N , splitting the average according to $i = j$ and $i \neq j$ gives

$$\mathbb{E}_0 \left[\sum_i H_{i,\Pi_N(i)} \sum_j K_{j,\Pi_N(j)} \right] = \frac{1}{N-1} \langle H, K \rangle_F. \quad (\text{E.4})$$

Indeed, the diagonal contribution is $N^{-1} \langle H, K \rangle_F$, while the zero-sum identities reduce the off-diagonal numerator to $\langle H, K \rangle_F$, with probability factor $1/[N(N-1)]$. Consequently,

$$G_N^\Pi(f, g) = \frac{4}{N-1} \left\langle H^{(N)}(f), H^{(N)}(g) \right\rangle_F, \quad I_N^\Pi(f) = \frac{4}{N-1} \|H^{(N)}(f)\|_F^2. \quad (\text{E.5})$$

Jensen's inequality and $\sum_i d_i^{(N)} = N$ imply $\|\mathcal{O}_N a\|_{\ell^2} \leq \sqrt{N} \|a\|_{L^2}$. Hence

$$\|H^{(N)}(f)\|_F \leq N \|f\|_{\mathcal{X}}, \quad 0 \leq \frac{I_N^\Pi(f)}{N} \leq \frac{4N}{N-1} \|f\|_{\mathcal{X}}^2 \leq 8 \|f\|_{\mathcal{X}}^2. \quad (\text{E.6})$$

This is the uniform Hilbert–Schmidt bound used below.

The limit of the rank Fisher form is more precise. Let $(\ell_m)_{m \geq 1}$ be an orthonormal shifted-Legendre basis of H and put $\tilde{\mathcal{O}}_N = N^{-1/2} \mathcal{O}_N$. The positive operator $\tilde{\mathcal{O}}_N^* \tilde{\mathcal{O}}_N$ is the Bernstein–Durrmeyer operator of degree $N-1$. It is triangular on the nested polynomial spaces: the beta-integral formula applied to a monomial of degree $m \leq N-1$ gives diagonal coefficient

$$\frac{N!(N-1)!}{(N+m)!(N-1-m)!} = \prod_{r=1}^m \frac{N-r}{N+r}.$$

Self-adjointness then makes the orthogonal differences between successive polynomial spaces invariant, so the ℓ_m are eigenfunctions, with

$$\lambda_{N-1,m} = \prod_{r=1}^m \frac{N-r}{N+r} \quad (1 \leq m \leq N-1), \quad \lambda_{N-1,m} = 0 \quad (m \geq N). \quad (\text{E.7})$$

For each fixed m , these eigenvalues lie in $[0, 1]$ and converge to one. If $f = \sum_{j,k \geq 1} c_{jk} \ell_j \otimes \ell_k$, then

$$\frac{I_N^\Pi(f)}{N} = \frac{4N}{N-1} \sum_{j,k \geq 1} \lambda_{N-1,j} \lambda_{N-1,k} |c_{jk}|^2. \quad (\text{E.8})$$

Dominated convergence in the square-summable coefficient array yields

$$\boxed{\frac{I_N^\Pi(f)}{N} \longrightarrow 4\|f\|_{\mathcal{X}}^2 \quad (f \in \mathcal{X}).} \quad (\text{E.9})$$

We now isolate the second channel. Conditional expectation of scores under $\Pi_N \mapsto [P_{\Pi_N}]$ and the law of total variance give

$$\Delta_N(f, g) := G_N^\Pi(f, g) - G_{[P]}^{(N)}(f, g) = \mathbb{E}_0[\text{Cov}_0(S_N^\Pi(f), S_N^\Pi(g) \mid [P_{\Pi_N}])] . \quad (\text{E.10})$$

Thus Δ_N is positive semidefinite and $0 \leq \Delta_N(f, f) \leq I_N^\Pi(f)$.

First suppose that $f \in \mathcal{X}_{\text{sym}}$ has finite rank. Its spectral decomposition has the form $f = \sum_{r=1}^R \alpha_r a_r \otimes a_r$, with centered orthonormal $a_r \in H$. Put $x_i = (\mathcal{O}_N a)_i$ for one fixed profile. The spectral convergence above and a bounded-function approximation give

$$\frac{1}{N} \sum_{i=1}^N x_i^2 \longrightarrow \|a\|_2^2, \quad \max_i |x_i| = o(\sqrt{N}), \quad \sum_i x_i = 0. \quad (\text{E.11})$$

For the second assertion, choose bounded b close to a in L^2 . Since $0 \leq d_i^{(N)} \leq N$ and $\int d_i^{(N)} = 1$, uniformly in i ,

$$\frac{|(\mathcal{O}_N a)_i|}{\sqrt{N}} \leq \frac{\|b\|_\infty}{\sqrt{N}} + \|a - b\|_2. \quad (\text{E.12})$$

Taking $N \rightarrow \infty$ and then $b \rightarrow a$ proves (E.11).

Let $X_N(a) = \sum_i x_i x_{\Pi_N(i)}$, $S_2 = \sum_i x_i^2$, $S_4 = \sum_i x_i^4$, and $(N)_r = N(N-1)\cdots(N-r+1)$. Directly grouping the four indices by coincidence pattern gives, for $N \geq 4$,

$$\begin{aligned} \mathbb{E}_0[X_N(a)^4] &= \frac{S_4}{N} + \frac{4S_4^2}{(N)_2} + \frac{3(S_2^2 - S_4)^2}{(N)_2} \\ &\quad + \frac{6(2S_4 - S_2^2)^2}{(N)_3} + \frac{9(S_2^2 - 2S_4)^2}{(N)_4}. \end{aligned} \quad (\text{E.13})$$

Equation (E.11) gives $S_2 = O(N)$ and $S_4 \leq (\max_i |x_i|)^2 S_2 = o(N^2)$. Substitution in (E.13) yields $\mathbb{E}_0[X_N(a)^4] = o(N^3)$. Since $S_N^\Pi(f) = 2 \sum_{r=1}^R \alpha_r X_N(a_r)$, Minkowski's inequality in L^4 gives

$$\mathbb{E}_0[S_N^\Pi(f)^4] = o(N^3) \quad (f \text{ symmetric and of fixed finite rank}). \quad (\text{E.14})$$

We now identify exactly the high-probability event used for the quotient step. Let $\mathcal{G}_N$ be the event that the strong interval tree of Π_N has a prime root and that every child of the root is either a leaf or a twin, i.e. a linear node with two leaf children. This is not the event that the whole incomparability graph is prime: twins are allowed. The strong interval tree is equivalently the modular decomposition tree of the permutation graph (Bouvel et al., 2009, Remark 1); hence $\mathcal{G}_N$ is $[P_{\Pi_N}]$ -measurable, since the unlabeled poset determines its incomparability graph up to isomorphism. Theorem 2 of Bouvel et al. (2009), whose proof applies their Lemma 1 with $c = 1$, states exactly that the complement of this prime-root-with-leaves-or-twins event has probability $O(N^{-1})$. Thus there are finite constants C_{fib} and N_{fib} such that $\mathbb{P}_0(\mathcal{G}_N^c) \leq C_{\text{fib}}/N$ for $N \geq N_{\text{fib}}$; the source does not specify these constants and we do not sharpen them.

For completeness, we prove that this same event gives the required fiber, including the permitted twins. Fix $\pi \in \mathcal{G}_N$, let $B_1, \dots, B_m$ be the maximal strong blocks below the root, and write the

corresponding inflation as $\pi = \alpha[\tau_1, \dots, \tau_m]$. The root condition says that the incomparability graph of the quotient α is prime, while every B_s has size one or two and hence $\tau_s \in \{1, 12, 21\}$. If $[P_\sigma] = [P_\pi]$, a poset isomorphism carries the canonical maximal strong modules to one another, preserving their sizes and their induced poset types. Contracting them therefore gives isomorphic quotient posets. Gallai's uniqueness theorem for the two transitive orientations of a prime comparability graph (Gallai, 1967), followed by normalization of both linear orders by rank, forces the quotient permutation of σ to be either α or α^{-1} . In the first case each internal pattern is fixed: on a two-element block, 12 induces a chain and 21 an antichain, so an isomorphism cannot exchange them. In the second case the exact inverse-of-an-inflation formula applies; the only internal patterns 1, 12, 21 are all involutions, and the result is π^{-1} . Consequently

$$\mathbb{P}_0(\mathcal{G}_N^c) \leq \frac{C_{\text{fib}}}{N} \quad (N \geq N_{\text{fib}}), \quad \Gamma_{[P_{\Pi_N}]} = \{\Pi_N, \Pi_N^{-1}\} \quad \text{on } \mathcal{G}_N, \quad (\text{E.15})$$

with the set understood without multiplicity for involutions.

For symmetric f , (E.2) gives $H^{(N)}(f)^\top = H^{(N)}(f)$, and hence $S_N^\Pi(f)(\sigma^{-1}) = S_N^\Pi(f)(\sigma)$. The conditional variance in (E.10) therefore vanishes on $\mathcal{G}_N$. Cauchy–Schwarz and (E.14)–(E.15) imply

$$\begin{aligned} 0 \leq \Delta_N(f, f) &\leq \mathbb{E}_0[S_N^\Pi(f)^2 \mathbf{1}_{\mathcal{G}_N^c}] \\ &\leq \mathbb{P}_0(\mathcal{G}_N^c)^{1/2} \mathbb{E}_0[S_N^\Pi(f)^4]^{1/2} = o(N). \end{aligned} \quad (\text{E.16})$$

To remove the finite-rank restriction, set $\mathcal{L}_N(f, g) = \Delta_N(f, g)/N$ on $\mathcal{X}_{\text{sym}}$. Equations (E.6) and (E.10) give the uniform bound

$$0 \leq \mathcal{L}_N(f, f) \leq 8\|f\|_{\mathcal{X}}^2. \quad (\text{E.17})$$

Choose symmetric finite-rank $f_R \rightarrow f$ in Hilbert–Schmidt norm before taking $N \rightarrow \infty$. The triangle inequality for the seminorm induced by the positive form $\mathcal{L}_N$ gives

$$\sqrt{\mathcal{L}_N(f, f)} \leq \sqrt{\mathcal{L}_N(f_R, f_R)} + \sqrt{8}\|f - f_R\|_{\mathcal{X}}. \quad (\text{E.18})$$

With R fixed, (E.16) makes the first term vanish as $N \rightarrow \infty$; only afterwards is $R \rightarrow \infty$ taken. Therefore

$$\boxed{\frac{\Delta_N(f, f)}{N} \rightarrow 0, \quad \frac{I_N^{[P]}(f)}{N} \rightarrow 4\|f\|_{\mathcal{X}}^2 \quad (f \in \mathcal{X}_{\text{sym}}).} \quad (\text{E.19})$$

For $0 \neq f \in \mathcal{X}_{\text{sym}}$, (E.9) provides a finite threshold $N_0(f)$ such that $I_N^\Pi(f) > 0$ for every $N \geq N_0(f)$. On that range the three ratios in (6.18) are well defined, and (E.9) and (E.19) give

$$\rho_N(f) \rightarrow 1, \quad \kappa_N(f) \rightarrow 1, \quad \eta_N^{\text{tot}}(f) \rightarrow 1. \quad (\text{E.20})$$

The threshold depends on f ; no ratio is assigned when its denominator is zero.

For the full tensor space, write $f = f_s + f_a$ according to (D.2). The transform in (E.2) intertwines coordinate swap with matrix transpose, so $H^{(N)}(f_s)$ is symmetric and $H^{(N)}(f_a)$ is skew-symmetric. Frobenius orthogonality in (E.5), together with the exact antisymmetric kernel in (D.5), gives

$$I_N^\Pi(f) = I_N^\Pi(f_s) + I_N^\Pi(f_a), \quad I_N^{[P]}(f) = I_N^{[P]}(f_s). \quad (\text{E.21})$$

Combining (E.9), (E.19), and (E.21),

$$\frac{I_N^{[P]}(f)}{4N\|f\|_{\mathcal{X}}^2} \rightarrow \frac{\|f_s\|_{\mathcal{X}}^2}{\|f_s\|_{\mathcal{X}}^2 + \|f_a\|_{\mathcal{X}}^2} \quad (f \neq 0). \quad (\text{E.22})$$

Finally, by the definition in (6.2),

$$\langle f, \widehat{F}_N f \rangle = \frac{I_N^{[P]}(f)}{4N} \longrightarrow \|\Pi_{\text{sym}} f\|_{\mathcal{X}}^2. \quad (\text{E.23})$$

Polarization gives weak-operator convergence to Π_{sym} . Since $0 \leq \widehat{F}_N \leq I$, also $\widehat{F}_N^2 \leq \widehat{F}_N$; using this inequality in the squared norm gives

$$\|\widehat{F}_N f - \Pi_{\text{sym}} f\|^2 \leq \langle f, \widehat{F}_N f \rangle + \|\Pi_{\text{sym}} f\|^2 - 2 \operatorname{Re} \langle \widehat{F}_N f, \Pi_{\text{sym}} f \rangle \longrightarrow 0.$$

This upgrades the weak convergence to

$$\boxed{\widehat{F}_N \xrightarrow{\text{SOT}} \Pi_{\text{sym}}}. \quad (\text{E.24})$$

This convergence is not uniform. For $h_N = p_N \otimes p_N / \|p_N\|_{L^2}^2$, one has $\|h_N\| = 1$, $h_N \perp V_N$, and therefore

$$\widehat{F}_N h_N = 0, \quad \Pi_{\text{sym}} h_N = h_N, \quad \|\widehat{F}_N - \Pi_{\text{sym}}\| \geq 1 \quad \text{for every } N. \quad (\text{E.25})$$

The result is an ambient Hilbert-space retention theorem. Its application to geometry is restricted to tangents already known to arise from admissible S1 paths.

F Exact matrices and spectra for $N = 2, 3, 4$

Put

$$x(t) := t - \frac{1}{2}, \quad q(t) := \left(t - \frac{1}{2}\right)^2 - \frac{1}{12}, \quad r(t) := \left(t - \frac{1}{2}\right)^3 - \frac{3}{20} \left(t - \frac{1}{2}\right). \quad (\text{F.1})$$

Direct integration gives the mutually orthogonal modal norms

$$\|x\|^2 = \frac{1}{12}, \quad \|q\|^2 = \frac{1}{180}, \quad \|r\|^2 = \frac{1}{2800}. \quad (\text{F.2})$$

For cardinality N , the continuous reference experiment consists of N iid copula observations. By (A.4), its score is $2 \sum_{k=1}^N f(U_k, V_k)$, and therefore

$$G_{\text{full}}^{(N)}(f, g) = 4N \langle f, g \rangle. \quad (\text{F.3})$$

In any basis of V_N , the nonzero eigenvalues of $\widehat{F}_N$ are the generalized eigenvalues of

$$G_{[P]}^{(N)} v = \lambda G_{\text{full}}^{(N)} v. \quad (\text{F.4})$$

So the matrix of the Fisher form in a non-normalized basis is not the matrix of the operator $\widehat{F}_N$; the continuous Gram matrix in (F.4) supplies the metric with respect to which the operator is represented.

Cardinality $N = 2$. Let $e_{11} = x \otimes x$. Formula (6.6), equivalently (A.13) specialized to the two poset classes, gives

$$G_{[P]}^{(2)}(f, g) = 256 \langle f, e_{11} \rangle \langle g, e_{11} \rangle. \quad (\text{F.5})$$

Since $\|e_{11}\|^2 = \|x\|^4 = 1/144$, evaluation of (F.5) on e_{11} , together with (F.3), yields

$$[G_{[P]}^{(2)}] = \left(\frac{1}{81}\right), \quad [G_{\text{full}}^{(2)}] = \left(\frac{1}{18}\right). \quad (\text{F.6})$$

Their generalized quotient is therefore

$$\text{spec}_+(\widehat{F}_2) = \left\{ \frac{1/81}{1/18} \right\} = \left\{ \frac{2}{9} \right\}. \quad (\text{F.7})$$

Cardinality $N = 3$. Use the orthogonal ordered basis

$$e_{11} = x \otimes x, \quad e_{12} = x \otimes q + q \otimes x, \quad e_{22} = q \otimes q. \quad (\text{F.8})$$

Substitution of the five class derivatives in the Fisher formula (A.13) gives the first matrix below. For the second, (F.2)–(F.3) give $12\|e_{11}\|^2 = 1/12$, $12\|e_{12}\|^2 = 12 \cdot 2\|x\|^2\|q\|^2 = 1/90$, and $12\|e_{22}\|^2 = 1/2700$. Hence

$$[G_{[P]}^{(3)}] = \text{diag} \left(\frac{1}{32}, \frac{1}{1200}, \frac{1}{180000} \right), \quad [G_{\text{full}}^{(3)}] = \text{diag} \left(\frac{1}{12}, \frac{1}{90}, \frac{1}{2700} \right). \quad (\text{F.9})$$

Both matrices are diagonal in the same basis, so their entrywise generalized quotients give

$$\text{spec}_+(\widehat{F}_3) = \left\{ \frac{3}{8}, \frac{3}{40}, \frac{3}{200} \right\}, \quad (\text{F.10})$$

with eigenvectors e_{11}, e_{12}, e_{22} , respectively. This records only Fisher anisotropy inside the three-dimensional visible support fixed in §4.

Cardinality $N = 4$. In exactly the order used in (6.9), set

$$\begin{aligned} e_{11} &= x \otimes x, & e_{12} &= x \otimes q + q \otimes x, & e_{13} &= x \otimes r + r \otimes x, \\ e_{22} &= q \otimes q, & e_{23} &= q \otimes r + r \otimes q, & e_{33} &= r \otimes r. \end{aligned} \quad (\text{F.11})$$

The exact class-score calculation from (A.13) gives

$$[G_{[P]}^{(4)}] = \begin{pmatrix} 4/75 & 0 & 0 & 0 & 0 & 0 \\ 0 & 8/3375 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/55125 & 1/354375 & 0 & 1/38587500 \\ 0 & 0 & 1/354375 & 11/455625 & 0 & -1/49612500 \\ 0 & 0 & 0 & 0 & 2/4134375 & 0 \\ 0 & 0 & 1/38587500 & -1/49612500 & 0 & 11/5402250000 \end{pmatrix}. \quad (\text{F.12})$$

Orthogonality and (F.2)–(F.3) give the corresponding continuous Gram matrix

$$[G_{\text{full}}^{(4)}] = \text{diag} \left(\frac{1}{9}, \frac{2}{135}, \frac{1}{1050}, \frac{1}{2025}, \frac{1}{15750}, \frac{1}{490000} \right). \quad (\text{F.13})$$

The isolated diagonal entries of (F.12)–(F.13) immediately produce the three pure channels

$$\lambda_{11} = \frac{4/75}{1/9} = \frac{12}{25}, \quad \lambda_{12} = \frac{8/3375}{2/135} = \frac{4}{25}, \quad \lambda_{23} = \frac{2/4134375}{1/15750} = \frac{4}{525}. \quad (\text{F.14})$$

On the remaining invariant block $\text{span}\{e_{13}, e_{22}, e_{33}\}$, taking $\det(G_{[P],\text{mix}}^{(4)} - \lambda G_{\text{full},\text{mix}}^{(4)})$ gives, up to a nonzero rational factor,

$$144703125\lambda^3 - 9975000\lambda^2 + 142000\lambda - 128. \quad (\text{F.15})$$

Its three roots are positive because the Fisher operator F_4 is positive definite on its support V_4 . For orientation only, as in §6, they are

$$0.0494521212879\dots, \quad 0.0185160720400\dots, \quad 0.000966047034941\dots \quad (\text{F.16})$$

Combining the three pure factors with the mixed block gives the full generalized characteristic determinant, again up to a nonzero rational factor:

$$(25\lambda - 12)(25\lambda - 4)(525\lambda - 4) (144703125\lambda^3 - 9975000\lambda^2 + 142000\lambda - 128). \quad (\text{F.17})$$

This appendix only documents the exact fixed- N calculations used in §6. It provides no all- N spectral formula, proves no monotonicity of eigenvalues with N , and does not identify F_N with P_N^{vis} . The generalized quotients compare the unlabeled-poset law with N continuous observations of the reference copula. They are not fractions of geometry and are not reconstruction results. No new claim is added here.

G Second-order derivatives and uniform-deletion kernel

The witness and the finite likelihood. With ℓ_1, ℓ_2 as in (7.1) and the generator ψ of (7.1)–(7.2),

$$\bar{\psi} = 0, \quad \|\psi\|_{L^2(D)}^2 = 2,$$

already established in §7; realizability is not reopened here. For exact polynomial integration it is convenient to use the equivalent factored form, obtained from (7.1) by direct expansion in u, v :

$$\psi(u, v) = -2\sqrt{15}(u - v)(6uv - 3u - 3v + 2). \quad (\text{G.1})$$

For $\pi \in S_N$ and two independent families of uniform order statistics $U_{(1)} < \dots < U_{(N)}$, $V_{(1)} < \dots < V_{(N)}$, set

$$T_\pi := \sum_{i=1}^N \psi(U_{(i)}, V_{(\pi(i))}). \quad (\text{G.2})$$

Since ψ is bounded on the compact D , the integrand $e^{2\varepsilon T_\pi}$ defining the numerator and the integrand $e^{2\varepsilon\psi}$ defining $Z(\varepsilon)$, together with their ε -derivatives of every order, are uniformly dominated on every compact interval of ε ; differentiation under the integral is therefore valid to every order, so both the numerator and $Z(\varepsilon)$ are real analytic, and $Z(\varepsilon) > 0$ for every $\varepsilon \in \mathbb{R}$ because the integrand is strictly positive. So, exactly as in (7.7),

$$\boxed{p_\pi(\varepsilon) = \frac{\langle e^{2\varepsilon T_\pi} \rangle_0}{N! Z(\varepsilon)^N},} \quad Z(\varepsilon) = \int_D e^{2\varepsilon\psi} d\mu_0, \quad (\text{G.3})$$

where $\langle \cdot \rangle_0$ is expectation, at $\varepsilon = 0$, under the two independent order-statistic vectors, and p_π is real analytic on all of $\mathbb{R}$.

First and second derivatives. Since $\bar{\psi} = 0$, $Z'(0) = 2 \int_D \psi d\mu_0 = 0$ and

$$Z''(0) = 4 \int_D \psi^2 d\mu_0 = 4\|\psi\|_{L^2(D)}^2.$$

Expanding $\langle e^{2\varepsilon T_\pi} \rangle_0 = 1 + 2\varepsilon \langle T_\pi \rangle_0 + 2\varepsilon^2 \langle T_\pi^2 \rangle_0 + O(\varepsilon^3)$ and $Z(\varepsilon)^{-N} = 1 - 2N\|\psi\|_{L^2(D)}^2 \varepsilon^2 + O(\varepsilon^4)$ in (G.3) and matching coefficients gives

$$\boxed{p'_\pi(0) = \frac{2}{N!} \langle T_\pi \rangle_0,} \quad \boxed{p''_\pi(0) = \frac{4}{N!} \left(\langle T_\pi^2 \rangle_0 - N\|\psi\|_{L^2(D)}^2 \right).} \quad (\text{G.4})$$

This reproduces (7.9); it is a scalar identity attached to the single path (G.3), not the definition of an operator on $\bigwedge^2 H$.

Parity. Since $\psi(v, u) = -\psi(u, v)$, relabeling the dummy variables in (G.2)–(G.3) — the U - and V -order-statistic families are independent and identically distributed, so exchanging their roles is a valid change of variables — turns T_π into $-T_{\pi^{-1}}$ after reindexing $j = \pi(i)$, and turns $Z(\varepsilon)$ into $Z(-\varepsilon)$ with $Z(-\varepsilon) = Z(\varepsilon)$. Hence

$$p_\pi(-\varepsilon) = p_{\pi^{-1}}(\varepsilon), \quad (\text{G.5})$$

which is (7.7). Each fiber Γ_C is closed under inversion (§3), so summing (G.5) over Γ_C gives

$$\boxed{\mu_{N,\varepsilon}^{[P]}(C) = \mu_{N,-\varepsilon}^{[P]}(C)} \quad (N \geq 2, C \in \mathcal{C}_N), \quad (\text{G.6})$$

which is (7.5). Every odd derivative of every class probability vanishes at $\varepsilon = 0$; this documents the computation of §7 and is not a new parity theorem.

Self-contained computation at $N = 2$. The two permutations of S_2 are the identity, whose fiber is the chain, and the transposition, whose fiber is the antichain:

$$T_{\text{chain}} = \psi(U_{(1)}, V_{(1)}) + \psi(U_{(2)}, V_{(2)}), \quad T_{\text{antichain}} = \psi(U_{(1)}, V_{(2)}) + \psi(U_{(2)}, V_{(1)}),$$

with $(U_{(1)}, U_{(2)})$ and $(V_{(1)}, V_{(2)})$ two independent pairs of order statistics of two uniforms on $[0, 1]$, jointly distributed with density 2 on each simplex $0 < t_1 < t_2 < 1$.

Both permutations of S_2 are involutions ($\text{id}^{-1} = \text{id}$, $\text{swap}^{-1} = \text{swap}$), so (G.5) gives $p_\pi(-\varepsilon) = p_\pi(\varepsilon)$ for each individually, and in particular

$$\langle T_{\text{chain}} \rangle_0 = \langle T_{\text{antichain}} \rangle_0 = 0, \quad (\text{G.7})$$

already at the level of a single permutation, not only of its class — reproducing exactly the vanishing recorded for $N = 2$ in (7.9).

For the second moments, expand $\psi = \ell_1 \otimes \ell_2 - \ell_2 \otimes \ell_1$ inside each square. Because the U -process is independent of the V -process, every cross term factors into a product of a two-point U -moment and a two-point V -moment (identically distributed to the U -moments). Write, for $j, k \in \{1, 2\}$ and $i \in \{1, 2\}$,

$$A_{jk} := \mathbb{E}[\ell_j(U_{(1)}) \ell_k(U_{(2)})], \quad M_i(jk) := \mathbb{E}[\ell_j(U_{(i)}) \ell_k(U_{(i)})]. \quad (\text{G.8})$$

Direct integration against the pair density 2 on $0 < t_1 < t_2 < 1$ and the marginal order-statistic densities $2(1 - t)$ ($i = 1$) and $2t$ ($i = 2$) gives the elementary values

$$A_{11} = A_{22} = 0, \quad A_{12} = -A_{21} = \frac{1}{\sqrt{15}}, \quad (\text{G.9})$$

$$M_1(11) = M_1(22) = M_2(11) = M_2(22) = 1, \quad M_1(12) = -\frac{2}{\sqrt{15}} = -M_2(12). \quad (\text{G.10})$$

Expanding $\psi(x, y)^2 = \ell_1(x)^2 \ell_2(y)^2 - 2\ell_1(x)\ell_2(x)\ell_1(y)\ell_2(y) + \ell_2(x)^2 \ell_1(y)^2$ and using independence of same- or different-index order statistics from the two processes,

$$\begin{aligned} \langle \psi(U_{(i)}, V_{(i)})^2 \rangle_0 &= 2M_i(11)M_i(22) - 2M_i(12)^2, \\ \langle \psi(U_{(1)}, V_{(2)})^2 \rangle_0 &= M_1(11)M_2(22) - 2M_1(12)M_2(12) + M_1(22)M_2(11), \end{aligned}$$

and, expanding the cross products $\psi(U_{(1)}, V_{(1)})\psi(U_{(2)}, V_{(2)})$ and $\psi(U_{(1)}, V_{(2)})\psi(U_{(2)}, V_{(1)})$ into four terms each and grouping U -factors against V -factors,

$$\langle \psi(U_{(1)}, V_{(1)})\psi(U_{(2)}, V_{(2)}) \rangle_0 = 2A_{11}A_{22} - 2A_{12}A_{21}, \quad \langle \psi(U_{(1)}, V_{(2)})\psi(U_{(2)}, V_{(1)}) \rangle_0 = -(A_{12}^2 + A_{21}^2).$$

Substituting (G.9)–(G.10): each diagonal term of T_{chain}^2 equals $2(1)(1) - 2(2/\sqrt{15})^2 = 22/15$, and the cross term equals $2(0)(0) - 2(1/\sqrt{15})(-1/\sqrt{15}) = 2/15$; each diagonal term of $T_{\text{antichain}}^2$ equals $1 - 2(-2/\sqrt{15})(2/\sqrt{15}) + 1 = 38/15$, and the cross term equals $-(1/15 + 1/15) = -2/15$. Summing,

$$\langle T_{\text{chain}}^2 \rangle_0 = 2 \cdot \frac{22}{15} + 2 \cdot \frac{2}{15} = \frac{16}{5}, \quad \langle T_{\text{antichain}}^2 \rangle_0 = 2 \cdot \frac{38}{15} + 2 \left(-\frac{2}{15} \right) = \frac{24}{5}. \quad (\text{G.11})$$

With $N\|\psi\|_{L^2(D)}^2 = 2 \cdot 2 = 4$, (G.4) and (G.11) give

$$\mu_2''(\text{antichain}) = \frac{4}{5} \left(\frac{24}{5} - 4 \right) = \frac{8}{5}, \quad \mu_2''(\text{chain}) = \frac{4}{5} \left(\frac{16}{5} - 4 \right) = -\frac{8}{5}, \quad (\text{G.12})$$

together with $\mu_{2,0}(\text{antichain}) = \mu_{2,0}(\text{chain}) = \frac{1}{2}$ from the uniform reference law on S_2 , and the exact check $\sum_C \mu_2''(C) = \frac{8}{5} - \frac{8}{5} = 0$, consistent with (G.6) summed over ε -independent total mass. These fractions reproduce (7.10); the derivation above is self-contained.

The uniform-deletion kernel. For $m \geq 3$ and unlabeled classes $C \in \mathcal{C}_m$, $D \in \mathcal{C}_{m-1}$, define

$$K_{m,m-1}(C, D) := \frac{1}{m} \# \{v \in C : [C \setminus \{v\}] = D\}. \quad (\text{G.13})$$

Well-defined on classes. Fix a labeled representative of C ; if $\phi : C \rightarrow C'$ is a poset isomorphism onto another representative, then for every $v \in C$ the restriction $\phi|_{C \setminus \{v\}}$ is an isomorphism $C \setminus \{v\} \rightarrow C' \setminus \{\phi(v)\}$, so ϕ carries $\{v \in C : [C \setminus v] = D\}$ bijectively onto $\{v' \in C' : [C' \setminus v'] = D\}$. The count in (G.13), hence $K_{m,m-1}(C, D)$, does not depend on the labeled representative chosen for C .

Markov kernel. Each summand is a nonnegative count, so $K_{m,m-1}(C, D) \geq 0$. Every $v \in C$ deletes to a poset lying in exactly one class $D \in \mathcal{C}_{m-1}$, so summing the count over all D counts each of the m elements of C exactly once:

$$\sum_{D \in \mathcal{C}_{m-1}} K_{m,m-1}(C, D) = \frac{1}{m} \sum_{v \in C} 1 = 1. \quad (\text{G.14})$$

Parameter-independence. (G.13) is a purely combinatorial count on unlabeled finite posets; it makes no reference to ε or to $\mu_{m,\varepsilon}^{[P]}$.

Deletion of an iid point. Let $X_1, \dots, X_m$ be iid from q_ε , and let V be uniform on $\{1, \dots, m\}$, independent of the sample. For every fixed j , $(X_i)_{i \neq j}$ is an iid sample of size $m-1$ from q_ε , being a fixed subvector of an iid vector; averaging over the uniform, independent choice of the deleted index leaves this law unchanged, so the joint law of $(X_i)_{i \neq V}$ is exactly the $(m-1)$ -fold product of q_ε . Passing to ranks and then to the unlabeled poset gives, coordinatewise,

$$\mu_{m-1,\varepsilon}^{[P]}(D) = \sum_{C \in \mathcal{C}_m} \mu_{m,\varepsilon}^{[P]}(C) K_{m,m-1}(C, D) \quad (D \in \mathcal{C}_{m-1}, m \geq 3). \quad (\text{G.15})$$

Composition to $N = 2$. As in (7.13), define

$$K_{N \rightarrow 2} := K_{3,2} \circ K_{4,3} \circ \dots \circ K_{N,N-1}, \quad K_{2 \rightarrow 2} := I. \quad (\text{G.16})$$

Iterating (G.15) from $m = N$ down to $m = 3$ gives, for every ε and every $N \geq 2$,

$$\mu_{2,\varepsilon}^{[P]} = K_{N \rightarrow 2} \mu_{N,\varepsilon}^{[P]}. \quad (\text{G.17})$$

Commutation with the jets. For fixed N , both sides of (G.17) are functions of ε valued in the finite-dimensional spaces $\mathbb{R}^{\mathcal{C}_2}$ and $\mathbb{R}^{\mathcal{C}_N}$, and $K_{N \rightarrow 2}$ is a fixed linear map between them, independent of ε by the third property of (G.13). Differentiating (G.17) k times therefore commutes termwise with this linear map — a matter of linearity on a finite-dimensional space, with no limit or asymptotic argument:

$$\boxed{(\mu_2^{[P]})^{(k)}(0) = K_{N \rightarrow 2}(\mu_N^{[P]})^{(k)}(0)} \quad (k \geq 1). \quad (\text{G.18})$$

Closure of Corollary 6. Assembling the pieces above: parity (G.6) gives $r_N(\gamma_\psi) \geq 2$ for every $N \geq 2$, by definition (7.8). If $(\mu_N^{[P]})''(0)$ vanished for some $N \geq 2$, (G.18) with $k = 2$ would force $K_{N \rightarrow 2}(\mu_N^{[P]})''(0) = 0$, i.e. $(\mu_2^{[P]})''(0) = 0$ — contradicting the exact values $(8/5, -8/5)$ of (G.12). Hence, exactly as in Corollary 6 of §7,

$$\boxed{r_N(\gamma_\psi) = 2 \quad \forall N \geq 2.} \quad (\text{G.19})$$

This closes that corollary; it is not presented as a distinct result.

This appendix treats a single explicit antisymmetric path, (G.1)–(G.3); it does not classify the general second differential on $\bigwedge^2 H$, does not define an operator Q_N , and does not define a quadratic null cone. It does not assert that every element of $\bigwedge^2 H$ has $r_N = 2$: (G.19) is an existence and non-vanishing statement for the single witness ψ of (7.1), propagated to every $N \geq 2$ by the parameter-independent deletion kernel of (G.13)–(G.18), not a classification result. No estimator, consistency statement, or rate is introduced, and no nonlinear reconstruction is claimed. The vanishing of the first-order jet in (G.6) is the exact isometric fold $\varepsilon \leftrightarrow -\varepsilon$ of Theorem 5 and (7.4)–(7.6); it must not be described as physical information loss. What (G.19) detects is the local magnitude of this deformation at second order, not its sign, which remains identified by the fold.

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